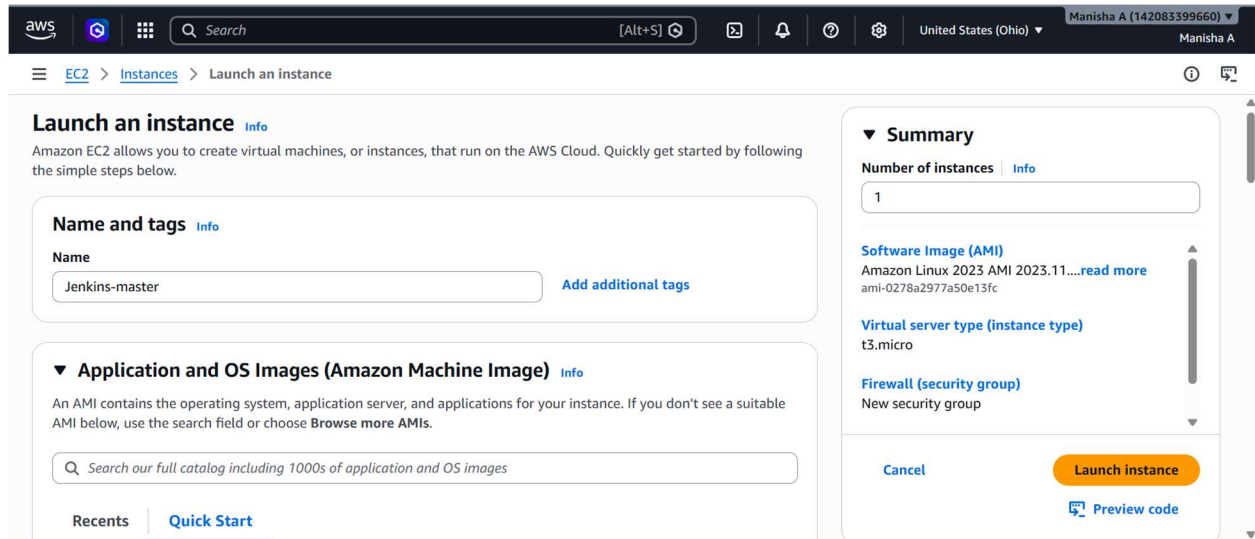


JENKINS-02

1. Configure 2 slave machines in the Jenkins master.

Lets create instance



Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name
Jenkins-master [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents [Quick Start](#)

Summary

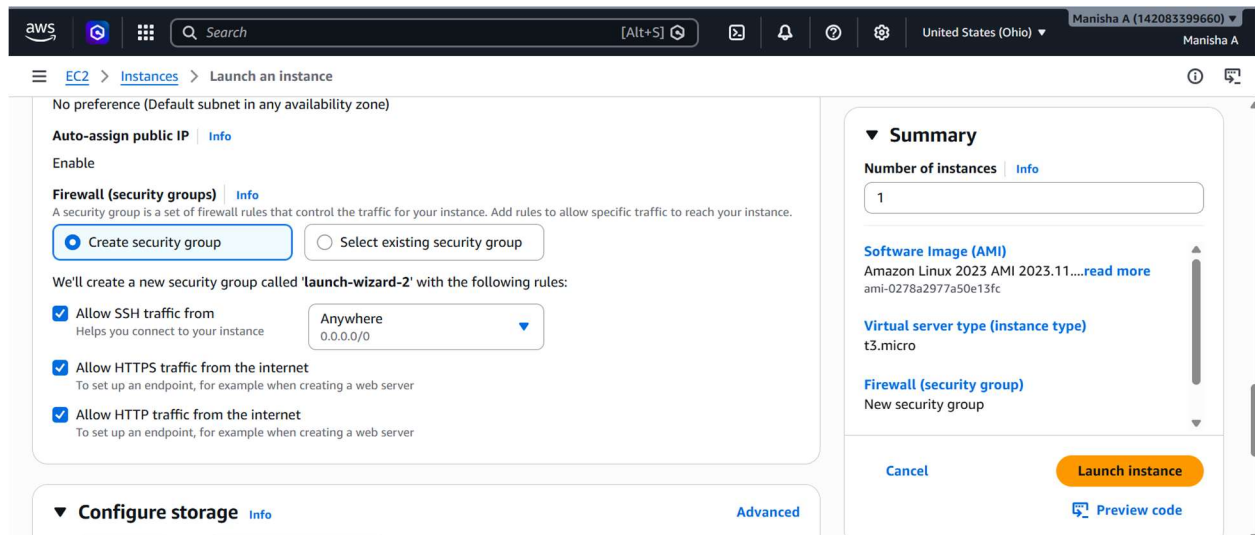
Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....[read more](#)
ami-0278a2977a50e13fc

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)



Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

- ☒ Allow SSH traffic from [Helps you connect to your instance](#)
- ☒ Allow HTTPS traffic from the internet [To set up an endpoint, for example when creating a web server](#)
- ☒ Allow HTTP traffic from the internet [To set up an endpoint, for example when creating a web server](#)

Configure storage [Info](#) [Advanced](#)

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....[read more](#)
ami-0278a2977a50e13fc

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

Configure Storage

1x 20 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐ S3 Files - new ☐ EFS ☐ FSx

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.11...[read more](#)
ami-0278a2977a50e13fc

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

Here we have created three instances

- Jenkins Master
- Slave-01
- Slave-02

Instances (1/3)

Save filter sets: Choose filter set. Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	slave-01	i-0a60af6086db37d71	Running	t3.micro	3/3 checks passed	View alarms
☐	slave-02	i-0a4f8de946f7da750	Running	t3.micro	3/3 checks passed	View alarms
☐	Jenkins-Master	i-09d416e616d472e73	Running	c7i-flex.large	3/3 checks passed	View alarms

Connect to Jenkins master

```
root@ip-172-31-11-90:~
manis@MANISHAREDDY MINGW64 ~/Downloads
$ ssh -i "Jenkins-1.pem" ec2-user@ec2-3-142-131-3.us-east-2.compute.amazonaws.com
The authenticity of host 'ec2-3-142-131-3.us-east-2.compute.amazonaws.com (3.142.131.3)' can't be established.
ED25519 key fingerprint is SHA256:SIck+vmic0t8C/bSbPR2a+k7uBc9WSbz7sG4181f2Lw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-142-131-3.us-east-2.compute.amazonaws.com' (ED25519) to the list of known hosts.

#_
~\#####_ Amazon Linux 2023
~~~\#####\
~~~\###|
~~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~~V~'~>
~~~
~~~.~.~
~~~/_/_/_
~~~/_/_/_
```

Install Jenkins

```
root@ip-172-31-11-90:~
[ec2-user@ip-172-31-11-90 ~]$ sudo su -
[root@ip-172-31-11-90 ~]# systemctl status jenkins
Unit jenkins.service could not be found.
[root@ip-172-31-11-90 ~]# sudo yum update
Amazon Linux 2023 Kernel Livepatch repository 347 kB/s | 38 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-11-90 ~]# sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-05-14 08:52:28-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.78.133, 2a04:4e42:83::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.78.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo 100%[=====] 267 --.-KB/s in 0s

2026-05-14 08:52:28 (9.20 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@ip-172-31-11-90 ~]# sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
```

```

root@ip-172-31-11-90:~
2026-05-14 08:52:28 (9.20 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@ip-172-31-11-90 ~]# sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
[root@ip-172-31-11-90 ~]# sudo yum upgrade
Jenkins-stable                               6.6 kB/s | 833 B      00:00
Jenkins-stable                               39 kB/s | 1.6 kB     00:00
Importing GPG key 0x14ABFC68:
  Userid      : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
  From        : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Is this ok [y/N]: y
Jenkins-stable                               16 kB/s | 2.1 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-11-90 ~]# sudo yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:00:12 ago on Thu May 14 08:52:56 2026.
Dependencies resolved.
=====
Package                                Arch    Version                                Repository    Size
=====
Installing:

```

```

root@ip-172-31-11-90:~
[root@ip-172-31-11-90 ~]# sudo yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:00:12 ago on Thu May 14 08:52:56 2026.
Dependencies resolved.
=====
Package                                Arch    Version                                Repository    Size
=====
Installing:
  java-21-amazon-corretto               x86_64  1:21.0.11+10-1.amzn2023.1            amazonlinux   218 k
Installing dependencies:
  alsa-lib                             x86_64  1.2.7.2-1.amzn2023.0.3                amazonlinux   503 k
  cairo                                x86_64  1.18.0-4.amzn2023.0.3                 amazonlinux   717 k
  dejavu-sans-fonts                     noarch  2.37-16.amzn2023.0.2                  amazonlinux   1.3 M
  dejavu-sans-mono-fonts                 noarch  2.37-16.amzn2023.0.2                  amazonlinux   467 k
  dejavu-serif-fonts                     noarch  2.37-16.amzn2023.0.2                  amazonlinux   1.0 M
  fontconfig                             x86_64  2.13.94-2.amzn2023.0.2                amazonlinux   273 k
  fonts-filesystem                       noarch  1:2.0.5-12.amzn2023.0.2               amazonlinux   9.5 k
  freetype                              x86_64  2.13.2-5.amzn2023.0.2                 amazonlinux   422 k
  giflib                                x86_64  5.2.1-9.amzn2023.0.3                  amazonlinux    48 k
  google-noto-fonts-common               noarch  20240401-1.amzn2023.0.2               amazonlinux    17 k
  google-noto-sans-vf-fonts              noarch  20240401-1.amzn2023.0.2               amazonlinux   593 k
  graphite2                             x86_64  1.3.14-7.amzn2023.0.2                 amazonlinux    97 k
  harfbuzz                               x86_64  7.0.0-2.amzn2023.0.2                  amazonlinux   873 k
  java-21-amazon-corretto-headless       x86_64  1:21.0.11+10-1.amzn2023.1            amazonlinux    97 M

```



```
root@ip-172-31-11-90:~
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-172-31-11-90 ~]# sudo yum install jenkins -y
Last metadata expiration check: 0:00:32 ago on Thu May 14 08:52:56 2026.
Dependencies resolved.
=====
Package                Architecture  Version      Repository    Size
=====
Installing:
jenkins                noarch       2.555.2-1    jenkins       95 M

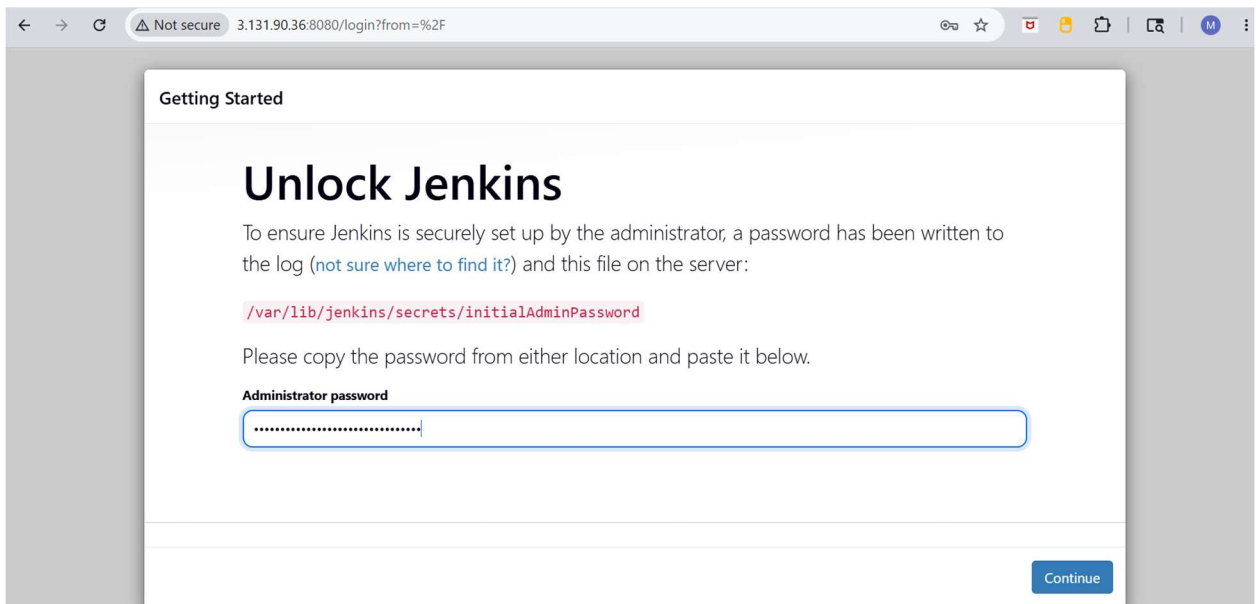
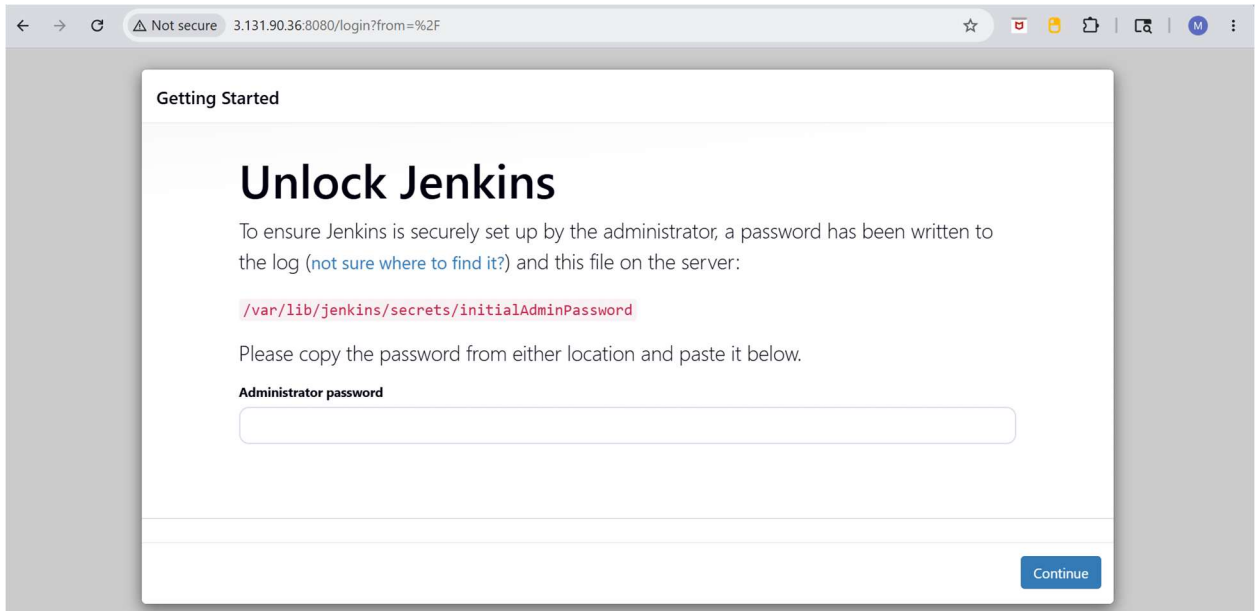
Transaction Summary
=====
Install 1 Package

Total download size: 95 M
Installed size: 95 M
Downloading Packages:
jenkins-2.555.2-1.noarch.rpm              72 MB/s | 95 MB    00:01
-----
Total                                      72 MB/s | 95 MB    00:01
Running transaction check
Transaction check succeeded.
```

```
root@ip-172-31-11-90:~
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Running scriptlet: jenkins-2.555.2-1.noarch 1/1
  Installing          : jenkins-2.555.2-1.noarch 1/1
  Running scriptlet: jenkins-2.555.2-1.noarch 1/1
  Verifying           : jenkins-2.555.2-1.noarch 1/1

Installed:
jenkins-2.555.2-1.noarch

Complete!
[root@ip-172-31-11-90 ~]# sudo systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@ip-172-31-11-90 ~]# sudo systemctl start jenkins
[root@ip-172-31-11-90 ~]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset:
   Active: active (running) since Thu 2026-05-14 08:53:59 UTC; 7s ago
   Main PID: 28177 (java)
```



Getting Started

Create First Admin User

Username

Password

Confirm password

Jenkins 2.555.1

[Skip and continue as admin](#) [Save and Continue](#)

Login to both the slaves through ssh

```
root@ip-172-31-37-9:~$ ssh -i "Jenkins-1.pem" ec2-user@ec2-13-59-174-119.us-east-2.compute.amazonaws.com
The authenticity of host 'ec2-13-59-174-119.us-east-2.compute.amazonaws.com (13.59.174.119)' can't be established.
ED25519 key fingerprint is SHA256:BwIztQAhiU7bgFR0ZrZPgKeT300jiAhd7ir+VZ7+aBU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-59-174-119.us-east-2.compute.amazonaws.com' (ED25519) to the list of known hosts.

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-37-9 ~]$ sudo su -
[root@ip-172-31-37-9 ~]# sudo yum install fontconfig java-21-amazon-corretto -y
```

```
root@ip-172-31-39-246:~$ ssh -i "Jenkins-1.pem" ec2-user@ec2-3-144-174-157.us-east-2.compute.amazonaws.com
The authenticity of host 'ec2-3-144-174-157.us-east-2.compute.amazonaws.com (3.144.174.157)' can't be established.
ED25519 key fingerprint is SHA256:0/BurtZF+X3W0A01kpFmRXSnXQuYeyEK7AfQJ4sh5No.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-144-174-157.us-east-2.compute.amazonaws.com' (ED25519) to the list of known hosts.

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-39-246 ~]$ sudo su -
```

Install java in both slaves

```
root@ip-172-31-37-9:~$ sudo su -
[root@ip-172-31-37-9 ~]# sudo yum install fontconfig java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 315 kB/s | 38 kB 00:00
Last metadata expiration check: 0:00:01 ago on Thu May 14 09:03:34 2026.
Dependencies resolved.

Package Arch Version Repository Size
Installing:
fontconfig x86_64 2.13.94-2.amzn2023.0.2 amazonlinux 273 k
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
Installing dependencies:
alsa-lib x86_64 1.2.7-2-1.amzn2023.0.3 amazonlinux 503 k
cairo x86_64 1.18.0-4.amzn2023.0.3 amazonlinux 717 k
dejavu-sans-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.3 M
dejavu-sans-mono-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 467 k
dejavu-serif-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.0 M
fonts-filesystem noarch 1:2.0.5-12.amzn2023.0.2 amazonlinux 9.5 k
freetype x86_64 2.13.2-5.amzn2023.0.2 amazonlinux 422 k
glib x86_64 5.2.1-9.amzn2023.0.3 amazonlinux 48 k
google-noto-fonts-common noarch 20240401-1.amzn2023.0.2 amazonlinux 17 k
google-noto-sans-vf-fonts noarch 20240401-1.amzn2023.0.2 amazonlinux 593 k
graphite2 x86_64 1.3.14-7.amzn2023.0.2 amazonlinux 97 k
harfbuzz x86_64 7.0.0-2.amzn2023.0.2 amazonlinux 873 k
```

```
root@ip-172-31-39-246:~$ sudo su -
[root@ip-172-31-39-246 ~]# sudo yum install fontconfig java-21-amazon-corretto -y
Amazon Linux 2023 Kernel Livepatch repository 315 kB/s | 38 kB 00:00
Dependencies resolved.

Package Arch Version Repository Size
Installing:
fontconfig x86_64 2.13.94-2.amzn2023.0.2 amazonlinux 273 k
java-21-amazon-corretto x86_64 1:21.0.11+10-1.amzn2023.1 amazonlinux 218 k
Installing dependencies:
alsa-lib x86_64 1.2.7-2-1.amzn2023.0.3 amazonlinux 503 k
cairo x86_64 1.18.0-4.amzn2023.0.3 amazonlinux 717 k
dejavu-sans-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.3 M
dejavu-sans-mono-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 467 k
dejavu-serif-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.0 M
fonts-filesystem noarch 1:2.0.5-12.amzn2023.0.2 amazonlinux 9.5 k
freetype x86_64 2.13.2-5.amzn2023.0.2 amazonlinux 422 k
glib x86_64 5.2.1-9.amzn2023.0.3 amazonlinux 48 k
google-noto-fonts-common noarch 20240401-1.amzn2023.0.2 amazonlinux 17 k
google-noto-sans-vf-fonts noarch 20240401-1.amzn2023.0.2 amazonlinux 593 k
graphite2 x86_64 1.3.14-7.amzn2023.0.2 amazonlinux 97 k
harfbuzz x86_64 7.0.0-2.amzn2023.0.2 amazonlinux 873 k
```

Install java packages

```
root@ip-172-31-37-9:~  
Complete!  
[root@ip-172-31-37-9 ~]# sudo yum install fontconfig java-21-openjdk -y  
Last metadata expiration check: 0:00:30 ago on Thu May 14 09:03:34 2026.  
Package fontconfig-2.13.94-2.amzn2023.0.2.x86_64 is already installed.  
No match for argument: java-21-openjdk  
Error: Unable to find a match: java-21-openjdk  
[root@ip-172-31-37-9 ~]# java -version  
openjdk version "21.0.11" 2026-04-21 LTS  
OpenJDK Runtime Environment Corretto-21.0.11.10.1 (build 21.0.11+10-LTS)  
OpenJDK 64-Bit Server VM Corretto-21.0.11.10.1 (build 21.0.11+10-LTS, mixed mode  
, sharing)  
[root@ip-172-31-37-9 ~]# rpm -qa | grep java-21  
java-21-amazon-corretto-headless-21.0.11+10-1.amzn2023.1.x86_64  
java-21-amazon-corretto-21.0.11+10-1.amzn2023.1.x86_64  
root@ip-172-31-39-246:~  
libpng-2.1.6-37-10.amzn2023.0.13.x86_64  
libxcb-1.17.0-1.amzn2023.0.1.x86_64  
pixman-0.43.4-1.amzn2023.0.4.x86_64  
xml-common-0.6.3-56.amzn2023.0.2.noarch  
Complete!  
[root@ip-172-31-39-246 ~]# java -version  
openjdk version "21.0.11" 2026-04-21 LTS  
OpenJDK Runtime Environment Corretto-21.0.11.10.1 (build 21.0.11+10-LTS)  
OpenJDK 64-Bit Server VM Corretto-21.0.11.10.1 (build 21.0.11+10-LTS, mixed mode  
, sharing)  
[root@ip-172-31-39-246 ~]# rpm -qa | grep java-21  
java-21-amazon-corretto-headless-21.0.11+10-1.amzn2023.1.x86_64  
java-21-amazon-corretto-21.0.11+10-1.amzn2023.1.x86_64
```

Create jenkins user

Create .ssh directory for jenkins user

```
root@ip-172-31-37-9:~  
, sharing)  
[root@ip-172-31-37-9 ~]# rpm -qa | grep java-21  
java-21-amazon-corretto-headless-21.0.11+10-1.amzn2023.1.x86_64  
java-21-amazon-corretto-21.0.11+10-1.amzn2023.1.x86_64  
[root@ip-172-31-37-9 ~]# useradd -m -s /bin/bash jenkins  
[root@ip-172-31-37-9 ~]# mkdir -p /home/jenkins/.ssh  
[root@ip-172-31-37-9 ~]# chown jenkins:jenkins /home/jenkins/.ssh  
[root@ip-172-31-37-9 ~]# chmod 700 /home/jenkins/.ssh  
[root@ip-172-31-37-9 ~]# ls  
root@ip-172-31-39-246:~  
OpenJDK 64-Bit Server VM Corretto-21.0.11.10.1 (build 21.0.11+10-LTS, mixed mode  
, sharing)  
[root@ip-172-31-39-246 ~]# rpm -qa | grep java-21  
java-21-amazon-corretto-headless-21.0.11+10-1.amzn2023.1.x86_64  
java-21-amazon-corretto-21.0.11+10-1.amzn2023.1.x86_64  
[root@ip-172-31-39-246 ~]# useradd -m -s /bin/bash jenkins  
[root@ip-172-31-39-246 ~]# mkdir -p /home/jenkins/.ssh  
[root@ip-172-31-39-246 ~]# chown jenkins:jenkins /home/jenkins/.ssh  
[root@ip-172-31-39-246 ~]# chmod 700 /home/jenkins/.ssh
```

Prepare the Jenkins Controller (Master) – we need to do in Jenkins-master server

Ensure the 'jenkins' user exists

The Jenkins RPM package normally creates this user automatically.

This command checks and creates it only if missing.

```
root@ip-172-31-11-90:~  
[root@ip-172-31-11-90 ~]# sudo su -  
Last login: Thu May 14 08:51:04 UTC 2026 on pts/1  
[root@ip-172-31-11-90 ~]# id -u jenkins &>/dev/null || useradd -m -s /bin/bash -  
d /var/lib/jenkins jenkins  
[root@ip-172-31-11-90 ~]# passwd jenkins  
Changing password for user jenkins.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
root@ip-172-31-11-90:~  
[root@ip-172-31-11-90 ~]# sudo usermod -s /bin/bash jenkins  
[root@ip-172-31-11-90 ~]# mkdir -p /var/lib/jenkins/.ssh  
[root@ip-172-31-11-90 ~]# chown jenkins:jenkins /var/lib/jenkins/.ssh  
[root@ip-172-31-11-90 ~]# chmod 700 /var/lib/jenkins/.ssh
```

Now switch to Jenkins user → Navigate to .ssh directory

```
[root@ip-172-31-11-90 ~]# su - jenkins  
[jenkins@ip-172-31-11-90 ~]$ cd ~/.ssh || mkdir ~/.ssh && cd ~/.ssh
```

Generate RSA key pair

```
[jenkins@ip-172-31-11-90 .ssh]$ ssh-keygen -t rsa -b 4096 -f id_rsa -q -N ""  
[jenkins@ip-172-31-11-90 .ssh]$ ls  
id_rsa id_rsa.pub
```


→ Set correct permissions

```
[jenkins@ip-172-31-11-90 .ssh]$ chmod 600 id_rsa
[jenkins@ip-172-31-11-90 .ssh]$ chmod 644 id_rsa.pub
[jenkins@ip-172-31-11-90 .ssh]$ ll
total 8
-rw----- 1 jenkins jenkins 3422 May 14 09:15 id_rsa
-rw-r--r-- 1 jenkins jenkins 776 May 14 09:15 id_rsa.pub
```

→ Display the public key (you will need it for slaves) → exit back to root

Public key: `/var/lib/jenkins/.ssh/id\_rsa.pub`

```
[jenkins@ip-172-31-11-90 .ssh]$ cat /var/lib/jenkins/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACQDCfVzqnR4pFN0U6IvJECJYa4kHdGiBkugMh7T7pg6S
Nj3bYkVraDKE6FSFkLG3PJrP1WGnSpANawOeoz6a5NUNo2qHxr/cULaYzr9PnEAD8ebJg86LPqtbP8Zm
FwHoUkX7rD1MrzDN8thIevPhHJNyJ8XVQjjPIUavG+T7jnQrgMUw0SWE9Uvce31prx6mg2Y1UFPzHxfA
oI7C7zKPYoM0GQSNyDVLZUH23Q2O2+vg54N8XgzezRyaLzuCPp8oeL1EAuAdmwxqiHUW2zQkCFjwb9E0
LVU71X5g1IAfsRko6yApVR0aztCyDdPECTXQ0hg3Eo2S+3vdxNZoA/MqqEz28D74/N8vL9wzyNS1YcpY
LAiNHeiKA7WSP3CdNkFLRJ1GzbkMnctQzZcVklx/2pG/Agnfa3xY2NJKsm0eh0YPaY/Co2ScP+7fmo74
K/bNjNAs0jjWjDf2aR2F5eTHkE2gG6iUrgnJhw0mtXCALgRifD5J7MFNb4eOYVtQOjUJG8fD1GXFpHbC
IP5Lo3S5Iaps0jk2SbkviBdcqt/a1I5qR0pMkN1K3jwdhUtxNuxWC6bw7fTWN14rTndp11/fg8q145sg
p1geDYVHVoQ1B36ncYG1NTYmROaaf6b2C8ELaGCTT/wvt2Vdn0WCxd6YlMT1SJ+6ASOFLFV9J6tjAeYZ
Rw== jenkins@ip-172-31-11-90.us-east-2.compute.internal
```

→ Private key: `/var/lib/jenkins/.ssh/id\_rsa`

```
root@ip-172-31-11-90:~
[jenkins@ip-172-31-11-90 .ssh]$ cat /var/lib/jenkins/.ssh/id_rsa
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzaC1rZXktZjEAAAAABG5vbmUAAAAAEbm9uZQAAAAAAAAABAAACFwAAAdzc2gtcn
NhAAAAAwEAAQAAAGeAwHVC6p0eKRTdFoILyRAiWGuJB3RogZLoDie0+6Y0kjY922JFa2gy
hohUhZCxtzyaz9Vhp0qQDWl jnqM+muTVDaNqh8a/3FC2mM6/T5xAA/Hm44P0iz6rWz/GZh
cB6FJF+6w9TK8wzflYSHlT4RyTcifF1UI4zyFGrxvk+450K4DFMNE1hPVL3Ht9aa8epoNm
NVBT8x8XwKcOwu8yj2KDDhKEjcg1S2VB9t0Njtv40eDfF4M3s0cmi87gj6fKH15RALgHZ
sMaoh1Fts0JAhY8G/RNC1V09V+YNSAH7EZK0sgKVUDGs7Qsg3TXAk10NIYNxKNKvt73cTW
aAPzKqhM9vA++PzFLy/cM8jUtWHKWCwIjYRIig01kj9wnTZYH0SdRs25DJ3LUM2XFZC8f9
qRvwIj32t8WNjSSrJtHodGD2mPwqNknD/u35q0+CV2zSZwOT041ow39mkdheXkx5BNoBuo
lK4JyYcNjRvVwgC4EYnw+SezBTW+HjmFbUD01CRvHw9RlxaR2wiD+S6N0uSGqbNI5Nkm5L4
gXXKrf2pSoakTqTJDZSt48HYVLcTbsVgum80301jZeK053aZZf34PKpeObIKdYHg2FR1aE
NQd+p3GBtTU2JkTmmn+m9gvBC2hgk0/8Fbd1XZ9FgsXemJTE9UifugEjhsXvfSerYwHmGU
cAAAdotvftdLVX7XQAAAAHC3NoLXJzYQAAAGeAwHVC6p0eKRTdFoILyRAiWGuJB3RogZLo
Die0+6Y0kjY922JFa2gyhohUhZCxtzyaz9Vhp0qQDWl jnqM+muTVDaNqh8a/3FC2mM6/T5
xAA/Hm44P0iz6rWz/GZhcB6FJF+6w9TK8wzflYSHlT4RyTcifF1UI4zyFGrxvk+450K4DF
MNE1hPVL3Ht9aa8epoNmNVBT8x8XwKcOwu8yj2KDDhKEjcg1S2VB9t0Njtv40eDfF4M3s
0cmi87gj6fKH15RALgHZsMaoh1Fts0JAhY8G/RNC1V09V+YNSAH7EZK0sgKVUDGs7Qsg3T
xAK10NIYNxKNKvt73cTWaAPzKqhM9vA++PzFLy/cM8jUtWHKWCwIjYRIig01kj9wnTZYH0
SdRs25DJ3LUM2XFZC8f9qRvwIj32t8WNjSSrJtHodGD2mPwqNknD/u35q0+CV2zSZwOT04
1ow39mkdheXkx5BNoBuo1K4JyYcNjRvVwgC4EYnw+SezBTW+HjmFbUD01CRvHw9RlxaR2wi
D+S6N0uSGqbNI5Nkm5L4gXXKrf2pSoakTqTJDZSt48HYVLcTbsVgum80301jZeK053aZZf
34PKpeObIKdYHg2FR1aENQd+p3GBtTU2JkTmmn+m9gvBC2hgk0/8Fbd1XZ9FgsXemJTE9U
ifugEjhsXvfSerYwHmGucAAADAQABAAACAEE7J1lU1oCENXVx66eDHBwx10g/8Na8+Ndx
Vb6cic9vUGsdIRkzGEJnCAerlPuFk+YYMhkrzf/3bviPqBjVjPArnw2aIG4AP2ejrAK
+5SSYQVEBEihJ6YsN5zovtrjw7EQ1qxxCr5zIQksT9MTvzumNwXTIMi6G1XLdKBPPQC/7D
wIvnejrQpLo83V/yUpENAW9eLWnEo/+tU6SSdKpJ7KfT4RKw0eTxmrnmPCyqPx3DHk1qDZ
Tn3KwEWN741xjtY+aq1SsAJyw2o0WHX/eHJPGd/45RHkMXsMVsws5yGLW+uBh0YZMOeuuH
Y6HYT/03WmKTFZCs+4jxJYspFEirat+Ujbewbhm6tcuAhDBI2fquddNRkeUuSFZ170JEIV
YCaSZySDTiAzWSHxmRHB0B7ZrYG9/jb6E4BpnXZ13PxxmceHBDcuqK1L6j8JP9agALaSp+
xEnMkJ/oAexhidYJPaPyR2DPRfnOa9eEqAJD25Hc51qjEGH8qIO5KpsKMjen+3uSHZVSea
st2x4a3cJlbcnSh7BScXVSPjdJECOI/WP966zIs7aJ9ECdwJGY4/kYQDS2xG1qkzn109kt
spHRZJ+CubUFqUXHokHAHpm5nmRroVqBTk1v5Pxg1ctT0Jj4i5Y5CnyQovnhFbfboLEuYM
7y4AFkBKoaodDj6SxtAAABAQCrcBB9kVgdpScXWDZSP+hX60J8zUH5Rhu1MrDB6kyu9ngn
yrLsLlYh9H/qR5Hygm/hf+dSARNnsxd9zavxtAWT3GtzeTg07g0CeI3dpehLp2r3EdL/okX
kKwksWGII6uGc/VKNTncELjUuMK1gEFH8etekY2Qqmk1cz+5Ga3q6fU03C56TGvrtUYePR9
7/iuLa1o2Zxqffq0mLTaRFAnDBxCYqaDuYBKH/EoZBu/8RRFO0quuDe+NUVHT1caVS81
41BWja42GP70S/M5nEQrtRSDHYovgtpeCnRe4FbcGRUDbbqoMbuvOCL16bwhCLsYkQN2L
```

```
[jenkins@ip-172-31-11-90 .ssh]$ exit
logout
```

Copy the Public Key to Both Agents

In both slave01 & slave02

```
[root@ip-172-31-37-9 ~]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACQCFVzqnR4p
NOU6IvJECJYa4kHdgiBkugMh7T7pg6SNj3bYkVradKE6FSFKLg3PjRPlWgnSpANawOeo26a5NUNO2qHx
r/cULayzr9PneAD8ebjg86LqtbP8ZmFwHouKx7rD1MrzDN8thiEvPHhJNyJ8XVQjJPIUavG+T7jnc
Mu0SW9Uvce31prx6mg2YIUFPzHxfAoI7C7zKPYoMGGQSNyDVLZUH23Q202+vg54N8XgzezRyaLzuCP
p8oel1EAUAdmwxq1HUW2zqCfjwb9EOLUv71X5g1IAfSRko6yApVR0aztcYDdPECTXQ0hg3Eo2S+3vdx
NZoA/Mqgcz28D74/N8VL9wzNS1YcpYLA1NNE1KA7NSP3cdNkFLR1JGzbKinctozZcVklx/2pg/AgnFa
3xY2NjKsm0eh0YPaY/co2ScP+7fmo74k/bNjNAs0jJwDf2aR2F5eThKE2g6iUrgnJhwOmtXCALqf
D5J7MFNB4e0YvtQ0jUJG8FD1GXFPbHbCIP5Lo3S5Iaps0jk2SbkviBdcqt/al15qR0pMkN1K3jwdhUtxN
uXwC6bw7FTwN14rTndp11/fg8q145sgp1geDYVHVQ1836ncYG1NTYmRoaf6b2C8ELaGCTT/wvt2Vdn
OWCxd6Y1MT1S3+6ASOFLV9J6tjAeyZRw== jenkins@ip-172-31-11-90.us-east-2.compute.in
ternal" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-37-9 ~]# chown jenkins:jenkins /home/jenkins/.ssh/authorized_key
s
[root@ip-172-31-37-9 ~]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-37-9 ~]#
```

```
[root@ip-172-31-39-246 ~]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACQCFVzqnR4
pFNOU6IvJECJYa4kHdgiBkugMh7T7pg6SNj3bYkVradKE6FSFKLg3PjRPlWgnSpANawOeo26a5NUNO2q
Hxr/cULayzr9PneAD8ebjg86LqtbP8ZmFwHouKx7rD1MrzDN8thiEvPHhJNyJ8XVQjJPIUavG+T7jnc
jrgMu0SW9Uvce31prx6mg2YIUFPzHxfAoI7C7zKPYoMGGQSNyDVLZUH23Q202+vg54N8XgzezRyaLzu
CPp8oel1EAUAdmwxq1HUW2zqCfjwb9EOLUv71X5g1IAfSRko6yApVR0aztcYDdPECTXQ0hg3Eo2S+3v
dxNZoA/Mqgcz28D74/N8VL9wzNS1YcpYLA1NNE1KA7NSP3cdNkFLR1JGzbKinctozZcVklx/2pg/Agn
Fa3xY2NjKsm0eh0YPaY/co2ScP+7fmo74k/bNjNAs0jJwDf2aR2F5eThKE2g6iUrgnJhwOmtXCALqf
1fD5J7MFNB4e0YvtQ0jUJG8FD1GXFPbHbCIP5Lo3S5Iaps0jk2SbkviBdcqt/al15qR0pMkN1K3jwdhU
txNuxwC6bw7FTwN14rTndp11/fg8q145sgp1geDYVHVQ1836ncYG1NTYmRoaf6b2C8ELaGCTT/wvt2V
dnOWCxd6Y1MT1S3+6ASOFLV9J6tjAeyZRw== jenkins@ip-172-31-11-90.us-east-2.compute.
internal" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-39-246 ~]# chown jenkins:jenkins /home/jenkins/.ssh/authorized_k
eys
[root@ip-172-31-39-246 ~]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-39-246 ~]#
```

Add Agents' Host Keys to known_hosts

Go to Jenkins server

Slave-1


```
[root@ip-172-31-11-90 ~]# ssh-keyscan -H 13.59.174.119 >> /var/lib/jenkins/.ssh/known_hosts
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
```

Slave-2

```
[root@ip-172-31-11-90 ~]# ssh-keyscan -H 13.59.174.119 >> /var/lib/jenkins/.ssh/known_hosts
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
# 13.59.174.119:22 SSH-2.0-OpenSSH_8.7
```

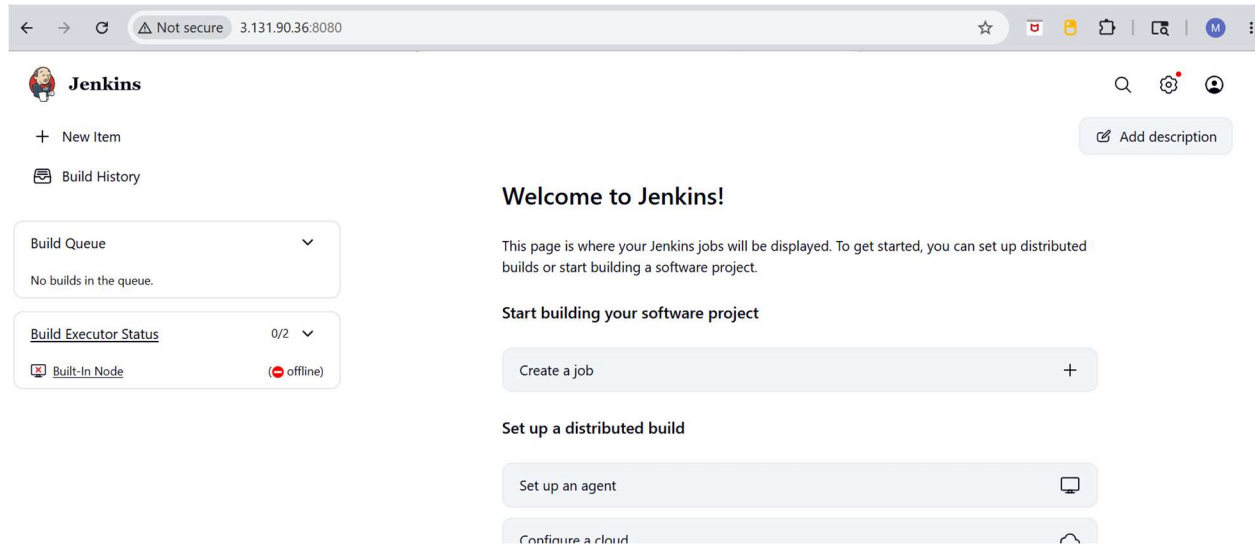
Fix ownership and permissions

```
[root@ip-172-31-11-90 ~]# chown jenkins:jenkins /var/lib/jenkins/.ssh/known_hosts
[root@ip-172-31-11-90 ~]# chmod 600 /var/lib/jenkins/.ssh/known_hosts
```

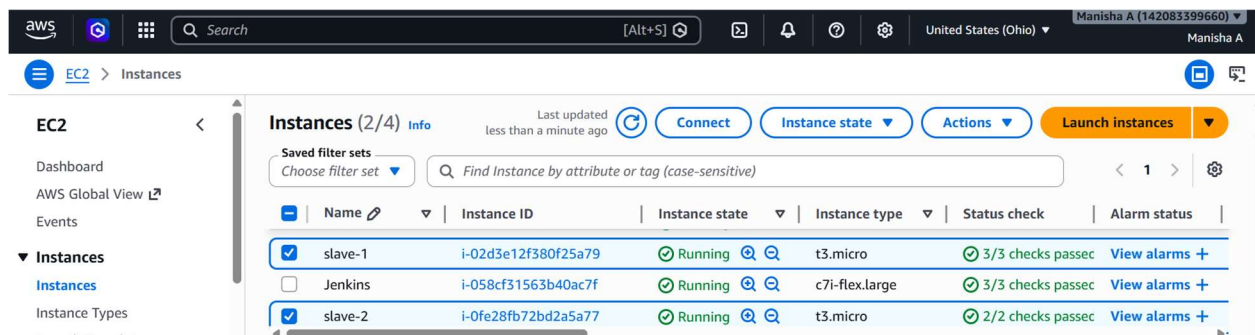
Test Password-less SSH Connectivity (From Master to Slave Machines)

```
[root@ip-172-31-11-90 ~]# sudo -u jenkins ssh jenkins@13.59.174.119
#_
~\#####_ Amazon Linux 2023
~~\#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'-'->
~~~
~~~.~.~
~~~/_m/'-'>
[jenkins@ip-172-31-37-9 ~]$ exit
logout
Connection to 13.59.174.119 closed.
[root@ip-172-31-11-90 ~]# sudo -u jenkins ssh jenkins@3.144.174.1573.144.174.157
ssh: Could not resolve hostname 3.144.174.1573.144.174.157: Name or service not known
[root@ip-172-31-11-90 ~]# sudo -u jenkins ssh jenkins@3.144.174.157
#_
~\#####_ Amazon Linux 2023
~~\#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'-'->
~~~
~~~.~.~
~~~/_m/'-'>
[jenkins@ip-172-31-39-246 ~]$ exit
logout
Connection to 3.144.174.157 closed.
```

We are able to login to Jenkins



Let's launch two slave instances



Connect them and install java


```
root@ip-172-31-46-251:/home/ec2-user
java-21-amazon-corretto-headless-1:21.0.11+10-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libpng-2:1.6.37-10.amzn2023.0.12.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
```

Slave-2

```
root@ip-172-31-35-113:/home/ec2-user
[ec2-user@ip-172-31-35-113 ~]$ sudo su
[root@ip-172-31-35-113 ec2-user]# yum update
Amazon Linux 2023 Kernel Livepatch repository 324 kB/s | 38 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-35-113 ec2-user]# yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:00:34 ago on Tue May 12 17:56:04 2026.
Dependencies resolved.

=====
Package                                Arch    Version                                Repository    Size
=====
Installing:
java-21-amazon-corretto                x86_64  1:21.0.11+10-1.amzn2023.1            amazonlinux   218 k
Installing dependencies:
alsa-lib                               x86_64  1.2.7.2-1.amzn2023.0.3               amazonlinux   503 k
cairo                                   x86_64  1.18.0-4.amzn2023.0.3                amazonlinux   717 k
dejavu-sans-fonts                       noarch  2.37-16.amzn2023.0.2                 amazonlinux   1.3 M
dejavu-sans-mono-fonts                 noarch  2.37-16.amzn2023.0.2                 amazonlinux   467 k
dejavu-serif-fonts                     noarch  2.37-16.amzn2023.0.2                 amazonlinux   1.0 M
fontconfig                             x86_64  2.13.94-2.amzn2023.0.2               amazonlinux   273 k
fonts-filesystem                       noarch  1:2.0.5-12.amzn2023.0.2              amazonlinux   9.5 k
freetype                               x86_64  2.13.2-5.amzn2023.0.2                amazonlinux   422 k
giflib                                  x86_64  5.2.1-9.amzn2023.0.3                 amazonlinux   48 k
```

```
root@ip-172-31-35-113:/home/ec2-user
java-21-amazon-corretto-headless-1:21.0.11+10-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libpng-2:1.6.37-10.amzn2023.0.12.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-172-31-35-113 ec2-user]# |
```

Create .ssh directory for jenkins user

Slave -1


```

[root@ip-172-31-46-251 ec2-user]# useradd -m -s /bin/bash jenkins
[root@ip-172-31-46-251 ec2-user]# sudo etc/passwd
sudo: etc/passwd: command not found
[root@ip-172-31-46-251 ec2-user]# sudo cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nolog
in
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
jenkins:x:1001:1001::/home/jenkins:/bin/bash
[root@ip-172-31-46-251 ec2-user]#

```

Created user in slave-2


```

root@ip-172-31-35-113:/home/ec2-user
[root@ip-172-31-35-113 ec2-user]# useradd -m -s /bin/bash jenkins
[root@ip-172-31-35-113 ec2-user]# sudo cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/sbin/nologin
dbus:x:81:81:System message bus:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993:/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72:/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
jenkins:x:1001:1001:/home/jenkins:/bin/bash
[root@ip-172-31-35-113 ec2-user]#

```

Created .ssh directory for jenkins user in slave1 & slave 2

```

root@ip-172-31-46-251:/home/ec2-user
[root@ip-172-31-46-251 ec2-user]# mkdir -p /home/jenkins/.ssh
[root@ip-172-31-46-251 ec2-user]# chown jenkins:jenkins /home/jenkins/.ssh
[root@ip-172-31-46-251 ec2-user]# chmod 700 /home/jenkins/.ssh
[root@ip-172-31-46-251 ec2-user]#

root@ip-172-31-35-113:/home/ec2-user
[root@ip-172-31-35-113 ec2-user]# mkdir -p /home/jenkins/.ssh
[root@ip-172-31-35-113 ec2-user]# chown jenkins:jenkins /home/jenkins/.ssh
[root@ip-172-31-35-113 ec2-user]# chmod 700 /home/jenkins/.ssh
[root@ip-172-31-35-113 ec2-user]#

```

I have connected to Jenkins masters

Ensure the 'jenkins' user exists

```

root@ip-172-31-46-5:~
[root@ip-172-31-46-5 ~]# id -u jenkins &>/dev/null || useradd -m -s /bin/bash -d /var/lib/jenkins jenkins
[root@ip-172-31-46-5 ~]# passwd jenkins
Changing password for user jenkins.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-46-5 ~]# sudo usermod -s /bin/bash jenkins
[root@ip-172-31-46-5 ~]#

```

Create .ssh directory for the jenkins user (with correct permissions)

```
[root@ip-172-31-46-5 ~]# mkdir -p /var/lib/jenkins/.ssh
[root@ip-172-31-46-5 ~]# chown jenkins:jenkins /var/lib/jenkins/.ssh
[root@ip-172-31-46-5 ~]# chmod 700 /var/lib/jenkins/.ssh
[root@ip-172-31-46-5 ~]#
```

Switch to Jenkins user and navigate to ssh directory

```
[root@ip-172-31-46-5 ~]# su - jenkins
[jenkins@ip-172-31-46-5 ~]$ cd .ssh
[jenkins@ip-172-31-46-5 .ssh]$
```

Generate RSA key pair

```
[jenkins@ip-172-31-46-5 .ssh]$ ssh-keygen -t rsa -b 4096 -f id_rsa -q -N ""
[jenkins@ip-172-31-46-5 .ssh]$
```

Set correct permissions

```
[jenkins@ip-172-31-46-5 .ssh]$ chmod 600 id_rsa
[jenkins@ip-172-31-46-5 .ssh]$ chmod 644 id_rsa.pub
[jenkins@ip-172-31-46-5 .ssh]$
```

Display the public key

cat /var/lib/jenkins/.ssh/id_rsa.pub

Switch back to root

Go to slave-1 & slave-2 → Add masters public key

```
root@ip-172-31-46-251:/home/ec2-user
[root@ip-172-31-46-251 ec2-user]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQCSPhwHP2pudm77vBRHPaB3LnbCintrUGH0oqSy3puFT2Qm0FyauRh9Nfcu1mMUNC57MC8BsYqSxy1Lxiu9HFOYUXvJ8IkiV0432PCNL6cuTzaxpS3kviP7/gkMAB9uJf6yZa9A083qVD1o3GrFqt0EupgY2cBcMoU
b7vy4NDZnnsY81MmWLLxy1cVjfdLa2HxUawCpHx3JPok+nuugF3856YxnqkvEjML3AUxgFK9ef/Xwnx
/tQISPq2QEBE3QzAH3SRkfGESANq+jzI701aL7qUGJ1A12S4dz/Ax+d-FBbMxKinMcYhzRs8Vv75ctnI
Uelczpym2QcUdExg/eFRqKzrYAKVhJEKEzP+c3eYVvwtZw5008LDPAbszvHHNH/95U7C6z2cYHcuXNF
fnh6ZFYMDv3y/07Hop35b7QFwa11MPCH91I9PmfM0uXx300u/6yT2r00BkY01gmb16HuekgaUAgpSx6
bv5s3gy2LIstX2i2WpIvoVots47FioHjQECuo/4U18aFz7/onhMHPpC7s1reSF809AXnv7M261sMH
QSPxFonVkwW3oSyJz3BKnoQP1WglFerwX8pHHUmuFxc85f8choTrstkB2mhH9wbUTQKYwAMfN+AsZolZ
tSPQwha08Pb+Q/a8P5TxjSQW141XGL6Br79x02apw== jenkins@ip-172-31-46-5.us-east-2.compu
te.internal" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-46-251 ec2-user]# chown jenkins:jenkins /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-46-251 ec2-user]# chmod 600 /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-46-251 ec2-user]#

root@ip-172-31-35-113:/home/ec2-user
[root@ip-172-31-35-113 ec2-user]# echo "ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQCSPhwHP2pudm77vBRHPaB3LnbCintrUGH0oqSy3puFT2Qm0FyauRh9Nfcu1mMUNC57MC8BsYqSxy1Lxiu9HFOYUXvJ8IkiV0432PCNL6cuTzaxpS3kviP7/gkMAB9uJf6yZa9A083qVD1o3GrFqt0EupgY2cBcMoU
b7vy4NDZnnsY81MmWLLxy1cVjfdLa2HxUawCpHx3JPok+nuugF3856YxnqkvEjML3AUxgFK9ef/Xwnx
/tQISPq2QEBE3QzAH3SRkfGESANq+jzI701aL7qUGJ1A12S4dz/Ax+d-FBbMxKinMcYhzRs8Vv75ctnI
Uelczpym2QcUdExg/eFRqKzrYAKVhJEKEzP+c3eYVvwtZw5008LDPAbszvHHNH/95U7C6z2cYHcuXNF
fnh6ZFYMDv3y/07Hop35b7QFwa11MPCH91I9PmfM0uXx300u/6yT2r00BkY01gmb16HuekgaUAgpSx6
bv5s3gy2LIstX2i2WpIvoVots47FioHjQECuo/4U18aFz7/onhMHPpC7s1reSF809AXnv7M261sMH
QSPxFonVkwW3oSyJz3BKnoQP1WglFerwX8pHHUmuFxc85f8choTrstkB2mhH9wbUTQKYwAMfN+AsZolZ
tSPQwha08Pb+Q/a8P5TxjSQW141XGL6Br79x02apw== jenkins@ip-172-31-46-5.us-east-2.compu
te.internal" >> /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-35-113 ec2-user]# chown jenkins:jenkins /home/jenkins/.ssh/authorized_keys
[root@ip-172-31-35-113 ec2-user]#
```

Add Agents' Host Keys to known_hosts

```
root@ip-172-31-46-251:/home/ec2-user
# 172.31.46.251:22 SSH-2.0-OpenSSH_8.7
[root@ip-172-31-46-251 ec2-user]# ssh-keyscan -H 3.144.142.89 >> /var/lib/jenkins/.ssh/known_hosts
# 3.144.142.89:22 SSH-2.0-OpenSSH_8.7
# 3.144.142.89:22 SSH-2.0-OpenSSH_8.7
# 3.144.142.89:22 SSH-2.0-OpenSSH_8.7
# 3.144.142.89:22 SSH-2.0-OpenSSH_8.7
# 3.144.142.89:22 SSH-2.0-OpenSSH_8.7
# 3.144.142.89:22 SSH-2.0-OpenSSH_8.7
[root@ip-172-31-46-251 ec2-user]#

root@ip-172-31-35-113:/home/ec2-user
# 18.225.176.229:22 SSH-2.0-OpenSSH_8.7
# 18.225.176.229:22 SSH-2.0-OpenSSH_8.7
# 18.225.176.229:22 SSH-2.0-OpenSSH_8.7
# 18.225.176.229:22 SSH-2.0-OpenSSH_8.7
# 18.225.176.229:22 SSH-2.0-OpenSSH_8.7
# 18.225.176.229:22 SSH-2.0-OpenSSH_8.7
[root@ip-172-31-35-113 ec2-user]#
```

Fix ownership and permissions

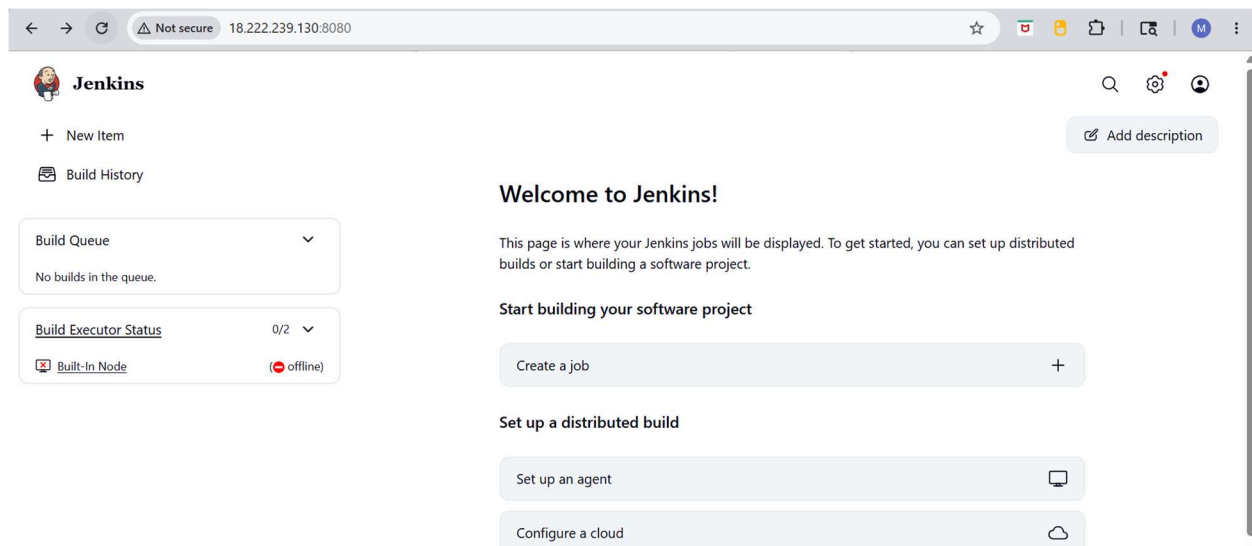
```
root@ip-172-31-46-251:/home/ec2-user
[root@ip-172-31-46-251 ec2-user]# chown jenkins:jenkins /var/lib/jenkins/.ssh/known_hosts
[root@ip-172-31-46-251 ec2-user]# chmod 600 /var/lib/jenkins/.ssh/known_hosts
[root@ip-172-31-46-251 ec2-user]#

root@ip-172-31-35-113:/home/ec2-user
[root@ip-172-31-35-113 ec2-user]# chown jenkins:jenkins /var/lib/jenkins/.ssh/known_hosts
[root@ip-172-31-35-113 ec2-user]# chmod 600 /var/lib/jenkins/.ssh/known_hosts
[root@ip-172-31-35-113 ec2-user]#
```

Slave-01

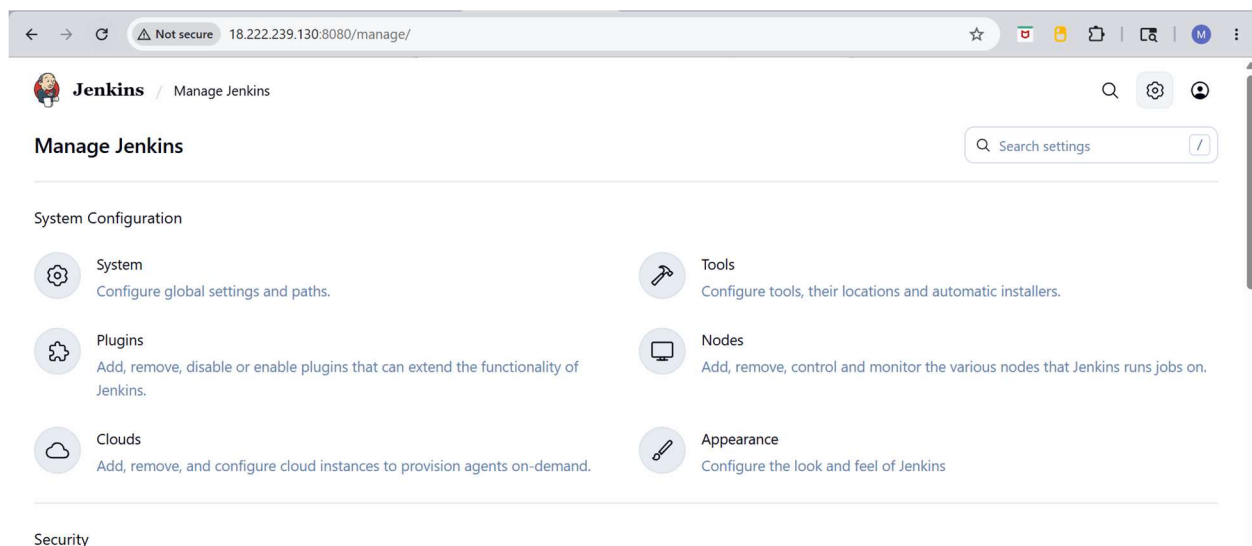
We are able to connect both the slaves from master

Login to Jenkins



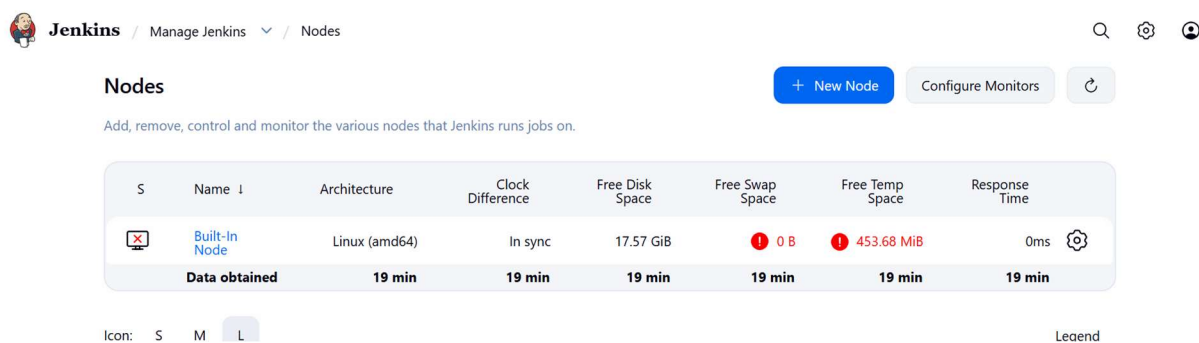
The image shows the Jenkins 'Welcome to Jenkins!' page. On the left sidebar, there are links for 'New Item', 'Build History', 'Build Queue' (showing 'No builds in the queue'), and 'Build Executor Status' (showing '0/2' and 'Built-In Node' as 'offline'). The main content area has a 'Welcome to Jenkins!' heading, a description of the page's purpose, and two sections: 'Start building your software project' with a 'Create a job' button, and 'Set up a distributed build' with buttons for 'Set up an agent' and 'Configure a cloud'.

Go to manage Jenkins → click on nodes



The image shows the 'Manage Jenkins' page. It features a search bar for settings. Under the 'System Configuration' section, there are six icons with labels: 'System' (Configure global settings and paths), 'Tools' (Configure tools, their locations and automatic installers), 'Plugins' (Add, remove, disable or enable plugins that can extend the functionality of Jenkins), 'Nodes' (Add, remove, control and monitor the various nodes that Jenkins runs jobs on), 'Clouds' (Add, remove, and configure cloud instances to provision agents on-demand), and 'Appearance' (Configure the look and feel of Jenkins). A 'Security' section is also visible at the bottom.

Click on new node



The image shows the 'Nodes' page in Jenkins. It includes a '+ New Node' button, a 'Configure Monitors' button, and a refresh icon. Below these is a table listing the nodes. The table has columns for 'S', 'Name', 'Architecture', 'Clock Difference', 'Free Disk Space', 'Free Swap Space', 'Free Temp Space', and 'Response Time'. There is one node listed: 'Built-In Node' (Linux (amd64)), which is 'In sync' and shows some resource usage (0 B Free Swap Space, 453.68 MiB Free Temp Space). At the bottom, there is a legend for icons (S, M, L) and a 'Legend' label.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	17.57 GiB	0 B	453.68 MiB	0ms
	Data obtained	19 min	19 min	19 min	19 min	19 min	19 min

Icon: S M L Legend



Jenkins / Manage Jenkins / Nodes / New node

Search Settings User

New node

Node name

slave-1

Type

☒ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

Select no.of executors and give remote root directory



Jenkins / Manage Jenkins / Nodes

Search Settings User

slave-01

Description ?

Plain text Preview

Number of executors ?

2

Remote root directory ?

/home/jenkins

Click launch method as launch agents via SSH → Click on add credentials → click on global → choose SSH username with private key

 Jenkins

Nodes / slave-1 / Configure



Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

3.144.142.89

Credentials ?

- none -

+ Add

The selected credentials cannot be found

Host Key Verification Strategy ?

Save

Apply

 Jenkins

Nodes / slave-1 / Configure



Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

3.144.142.89

Credentials ?

- none -

+ Add

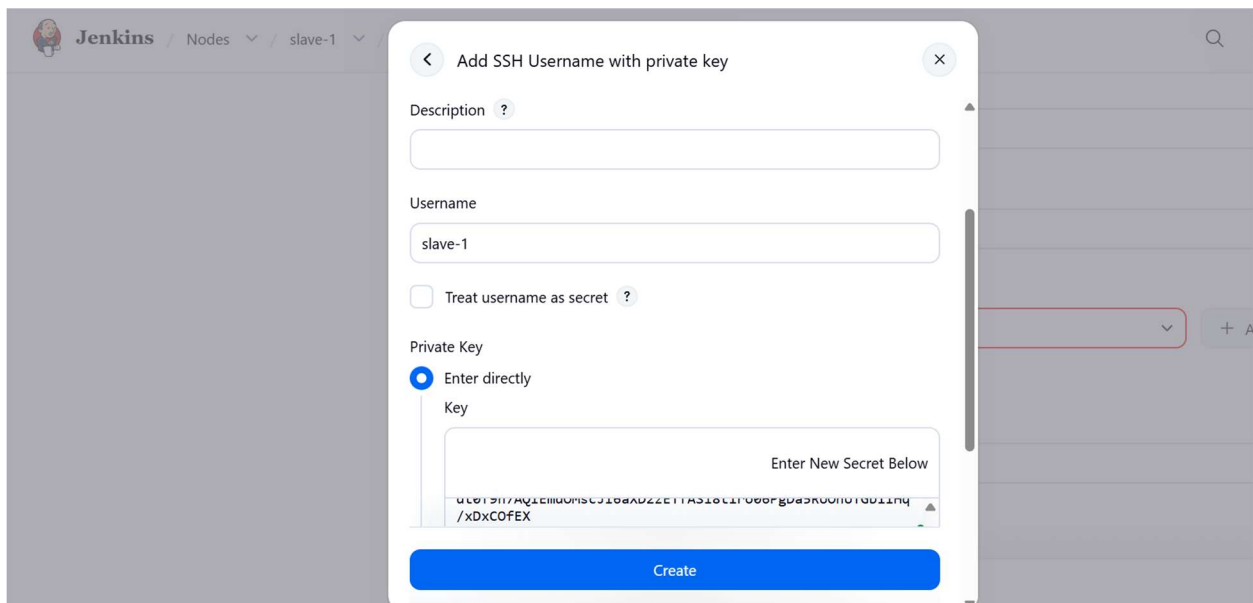
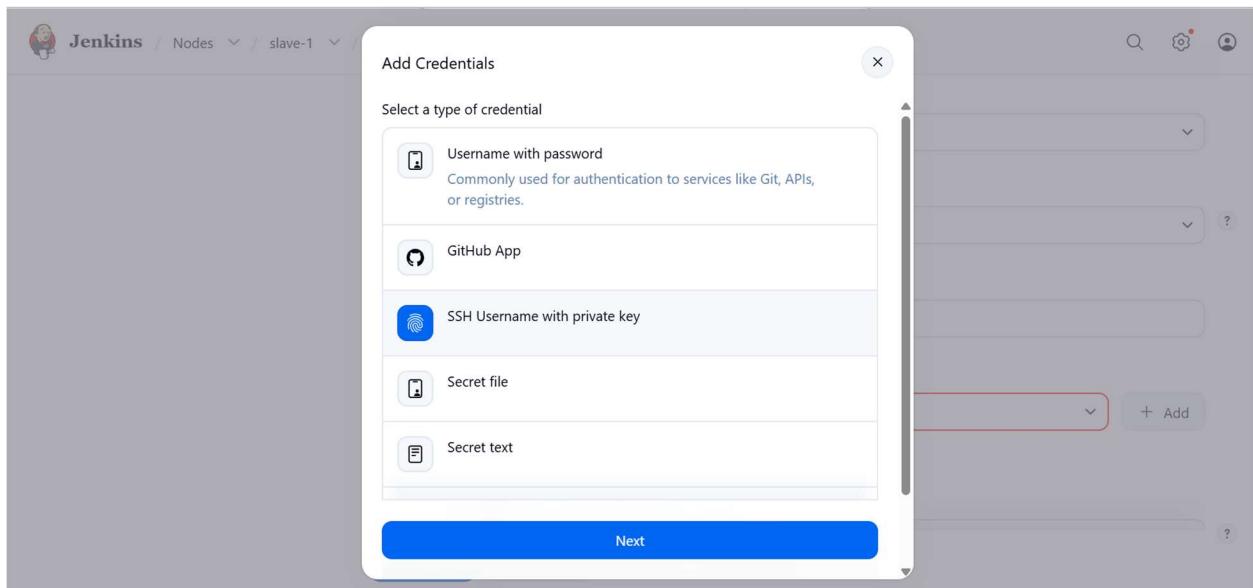
The selected credentials cannot be found

Host Key Verification Strategy ?

System

Global

Credentials that should be available everywhere.



Once after creating credentials → add it

 **Jenkins**

Nodes

▼

slave-01

▼

Configure







☐ Disable deferred wipeout on this node ?

☒ Disk Space Monitoring Thresholds

Free Disk Space Threshold ?

300MiB

Free Disk Space Warning Threshold ?

300MiB

Free Temp Space Threshold ?

300MiB

Free Temp Space Warning Threshold ?

300MiB

[illegible]

Jenkins / Manage Jenkins / Nodes

Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

+ New Node Configure Monitors

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	17.55 GiB	0 B	1.86 GiB	0ms
	slave-01		N/A	N/A	N/A	N/A	N/A
	Data obtained		58 min	58 min	58 min	58 min	58 min

Icon: S M L

- Delete Agent
- Configure
- Build History
- Load Statistics
- Script Console
- Log

Legend

It is connected successfully

Jenkins / Nodes / slave-01 / Log

```
Source agent hash is 7A9A77A64F02EB9F5DBCEFF9E9A349E2. Installed agent hash is 7A9A77A64F02EB9F5DBCEFF9E9A349E2
Verified agent jar. No update is necessary.
Expanded the channel window size to 4MB
[05/14/26 10:53:07] [SSH] Starting agent process: cd "/home/jenkins" && java -jar remoting.jar -workDir /home/jenkins -jar-cache /home/jenkins/remoting/jarCache
May 14, 2026 10:53:07 AM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/jenkins/remoting as a remoting work directory
May 14, 2026 10:53:07 AM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/jenkins/remoting
<===[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

Create a job

Click on Jenkins → new item

New Item

Enter an item name

job-1

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

General

Enabled 

Description

job-01

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterized ?

☐ Throttle builds ?

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterized ?

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

linux-01

Label linux-01 matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

 Jenkins

job-1

Configure

Search

Help

User

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See the list of available environment variables

pwd

ls

Advanced

Click on build now

 Jenkins

job-1

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

job-1

job-01

Permalinks

Builds >

Today

✓ #1 11:02 am

Jenkins / job-1 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

#1 (14-May-2026, 11:02:51 am)

Started by user [admin](#)

This run spent:

- 3 ms waiting;
- 0.39 sec build duration;
- 0.39 sec total from scheduled to completion.

No changes.

Add description

Keep this build forever

Started 39 sec ago

Took 0.39 sec on [slave-01](#)

Jenkins / job-1 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Console

Download

Copy

View as plain text

```
Started by user admin
Running as SYSTEM
Building remotely on slave-01 (linux-01) in workspace /home/jenkins/workspace/job-1
[job-1] $ /bin/sh -xe /tmp/jenkins16691721702031512531.sh
+ pwd
/home/jenkins/workspace/job-1
+ ls
Finished: SUCCESS
```

Same for slave 2

Jenkins / Manage Jenkins / Nodes / New node

New node

Node name

slave-02

Type



Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.



Copy Existing Node

Create



Jenkins

/ Manage Jenkins ▾ / Nodes ▾

slave-02

Description ?

Plain text [Preview](#)

Number of executors ?

1

Remote root directory ?

/home/jenkins

Save



Launch method ?

Launch agents via SSH

Host ?

3.144.174.157

Credentials ?

jenkins

Host Key Verification Strategy ?

Known hosts file Verification Strategy

Advanced ▾

Availability ?

Save



☐ Disable deferred wipeout on this node ?

☒ Disk Space Monitoring Thresholds

Free Disk Space Threshold ?

300Mib

Free Disk Space Warning Threshold ?

300Mib

Free Temp Space Threshold ?

300Mib

Free Temp Space Warning Threshold ?

300Mib

Save



```
dot "$@"
}
[05/14/26 11:13:32] [SSH] Starting sftp client.
[05/14/26 11:13:32] [SSH] Copying latest remoting.jar...
[05/14/26 11:13:32] [SSH] Copied 1,407,936 bytes.
Expanded the channel window size to 4MB
[05/14/26 11:13:32] [SSH] Starting agent process: cd "/home/jenkins" && java -jar remoting.jar -workDir
/home/jenkins -jar-cache /home/jenkins/remoting/jarCache
May 14, 2026 11:13:32 AM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using /home/jenkins/remoting as a remoting work directory
May 14, 2026 11:13:32 AM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to /home/jenkins/remoting
<==[JENKINS REMOTING CAPACITY]==>channel started
Remoting version: 3355.v388858a_47b_33
Launcher: SSHLauncher
Communication Protocol: Standard in/out
This is a Unix agent
Agent successfully connected and online
```

Create a job



New Item

Enter an item name

job-2

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK



Configure



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

General

Enabled



Description

job-2

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterized ?

☐ Throttle builds ?

Jenkins / job-2 / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Throttle builds ?

☐ Execute concurrent builds if necessary ?

☒ Restrict where this project can be run ?

Label Expression ?

linux-02

Label linux-02 matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾

Jenkins / job-2 / Configure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

ls

Jenkins / job-2 / Build Time Trend

Delete Project

Rename

Builds >

Filter

Today

✓ #1 11:17 am

mins

0

#1 slave-02

S	Build 1	Time Since	Duration	Agent	
✓	#1	2.6 sec	0.45 sec	slave-02	🗄

Icon: S M L

Jenkins / job-2 / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Console

Download Copy View as plain text

```
Started by user admin
Running as SYSTEM
Building remotely on slave-02 (linux-02) in workspace /home/jenkins/workspace/job-2
[job-2] $ /bin/sh -xe /tmp/jenkins10765219534367690347.sh
+ ls
Finished: SUCCESS
```

2. Configure webhooks to the Jenkins job

Go to manage Jenkins → click on plugins

Jenkins / Manage Jenkins

Manage Jenkins

Search settings

System Configuration

- System
Configure global settings and paths.
- Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Clouds
Add, remove, and configure cloud instances to provision agents on-demand.
- Tools
Configure tools, their locations and automatic installers.
- Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Appearance
Configure the look and feel of Jenkins

Security

Check whether the git plugins are installed

 Jenkins

Manage Jenkins

Plugins

5

Updates

Available plugins

Installed plugins

Advanced settings

git

Name	Health	Enabled
Git client plugin 6.6.0 Utility plugin for Git support in Jenkins. Report an issue with this plugin	100	☒
Git plugin 5.10.1 This plugin integrates Git with Jenkins. Report an issue with this plugin	100	☒
GitHub API Plugin 1.330-492.v3941a_032db_2a_ This plugin provides GitHub API for other plugins. Report an issue with this plugin	97	☒
GitHub Branch Source Plugin 1967.1969.v205fd594c821 Multibranch projects and organization folders from GitHub. Maintained by CloudBees, Inc. Report an issue with this plugin	97	☒

Create a new job

 Jenkins

+ New Item

Build History

Project Relationship

Check File Fingerprint

Build Queue

No builds in the queue.

Build Executor Status

0/2

Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

 Jenkins

New Item

Enter an item name

job-1

Select an item type

 Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

 Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Jenkins / job-1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled ☒

Description

webhook job

Plain text [Preview](#)

☐ Enable project-based security

☐ Discard old builds ?

☐ GitHub project

☐ This project is parameterized ?

Under source code management → click on git → provide repo url

Jenkins / job-1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/manisha-a99/Techie_horizon_Login_app.git

Credentials ?

- none -

+ Add

Advanced ▾

Jenkins / job-1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Branch Specifier (blank for 'any') ?

*/master

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Jenkins

job-1

Configure

Triggers

Environment

Build Steps

Post-build Actions

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐

Trigger builds remotely (e.g., from scripts) ?

☐

Build after other projects are built ?

☐

Build periodically ?☐☒☐

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☐

Save

Apply

Choose build step as execute shell

Jenkins

job-1

Configure

Triggers

Environment

Build Steps

Post-build Actions

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

+ Add build step

Filter

Execute Windows batch command

Execute shell

Invoke Ant

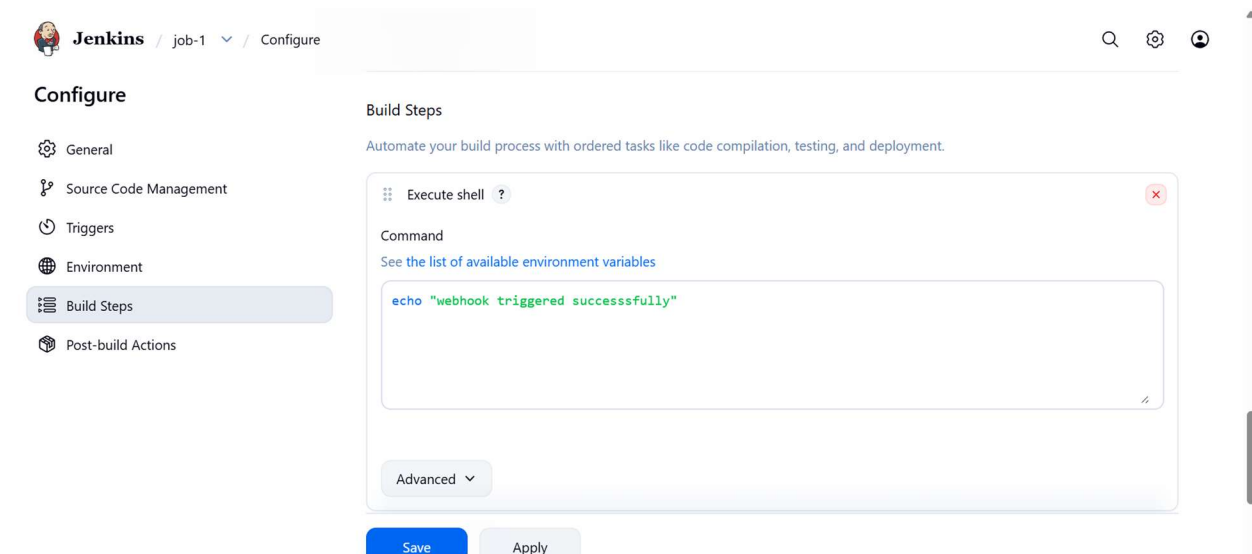
Invoke Gradle script

Invoke top-level Maven targets

Run with timeout

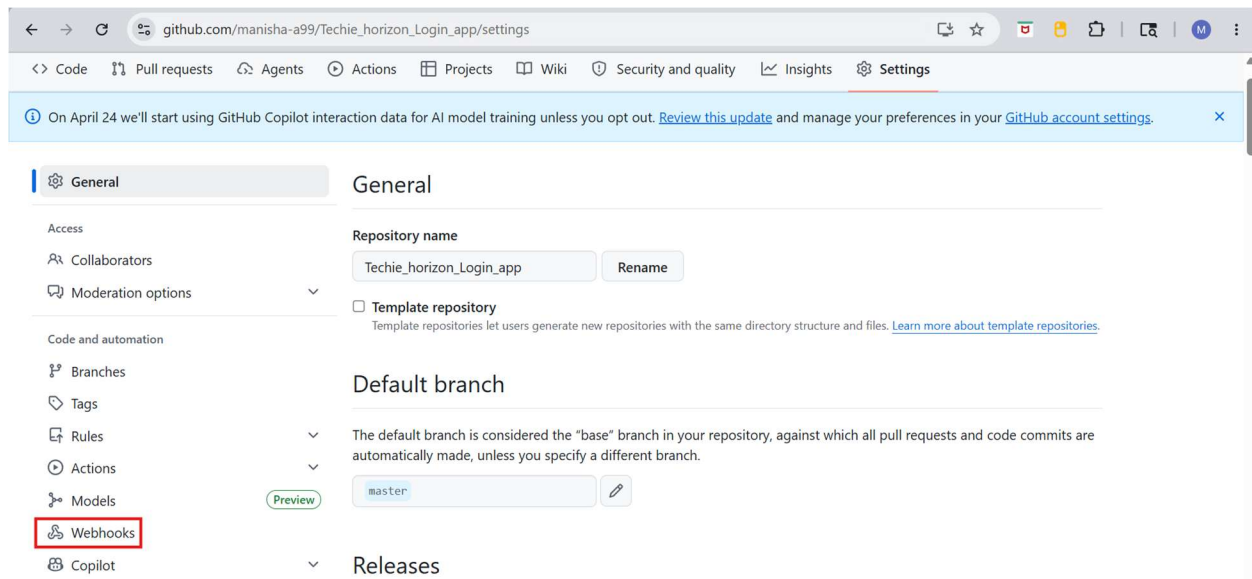
Set build status to "pending" on GitHub commit

... sending notifications, archiving artifacts, or triggering other jobs.



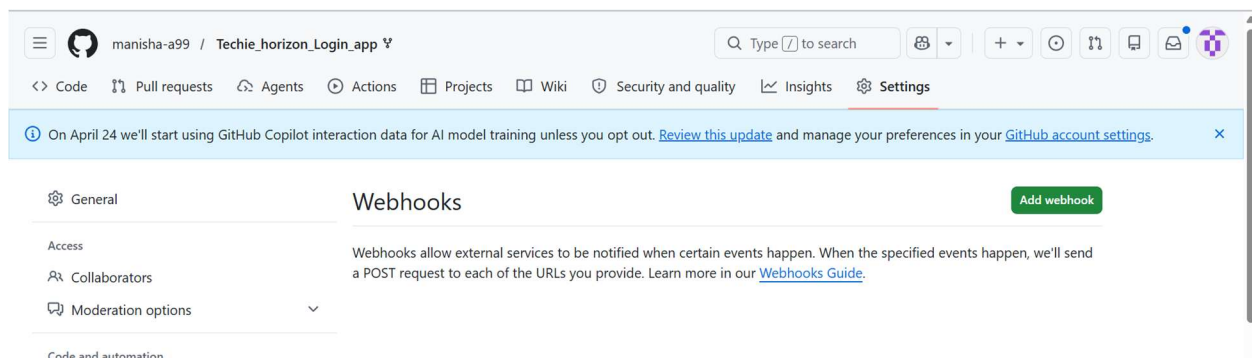
The image shows the Jenkins 'Configure' page for a job named 'job-1'. On the left sidebar, the 'Build Steps' option is selected. The main area is titled 'Build Steps' and contains a single step named 'Execute shell'. The command field for this step contains the text: `echo "webhook triggered successfully"`. Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration area are 'Save' and 'Apply' buttons.

Go to github repository settings and create webhooks



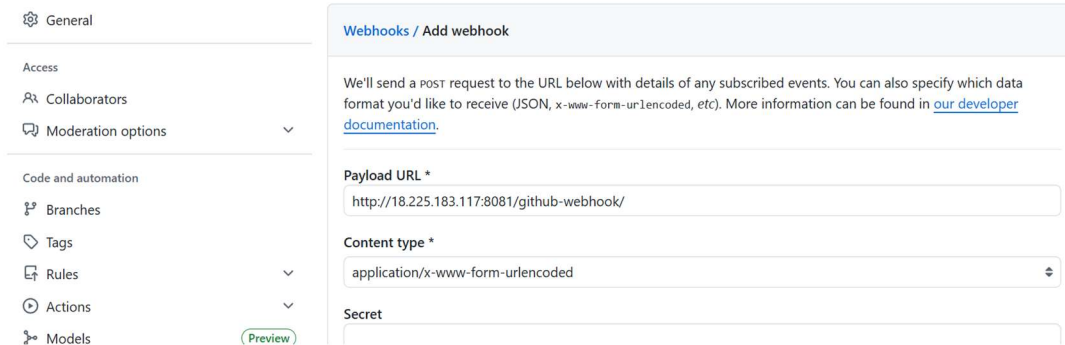
The image shows the GitHub repository settings page for 'Techie_horizon_Login_app'. The 'General' tab is selected in the left sidebar. The 'Webhooks' option in the sidebar is highlighted with a red box. The main content area shows the 'General' settings, including the repository name 'Techie_horizon_Login_app', a 'Rename' button, and a 'Template repository' checkbox. Below this is the 'Default branch' section, which shows 'master' as the default branch. The 'Releases' section is partially visible at the bottom.

Click on add webhook



The image shows the GitHub repository settings page for 'Techie_horizon_Login_app', specifically the 'Webhooks' tab. The 'Add webhook' button is visible in the top right corner. The main content area explains that webhooks allow external services to be notified when certain events happen, and it provides a link to the 'Webhooks Guide'.

Add payroll url → give public ip of Jenkins server, portnumber /github-webhook



General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks / Add webhook

We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in [our developer documentation](#).

Payload URL *

http://18.225.183.117:8081/github-webhook/

Content type *

application/x-www-form-urlencoded

Secret

Select on what event the build should trigger → click on add webhook

SSL verification

 By default, we verify SSL certificates when delivering payloads.

☐ Enable SSL verification ☒ **Disable (not recommended)**

Which events would you like to trigger this webhook?

- ☐ Just the push event.
- ☒ **Send me everything.**
- ☐ Let me select individual events.

☒ **Active**

We will deliver event details when this hook is triggered.

Update webhook

Delete webhook

Setup is successfully done

General
Access
Collaborators
Moderation options
Code and automation
Branches

Webhooks

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

✓ <http://18.225.183.117:8081/github-...> (all events)

Edit
Delete

Last delivery was successful.

Add webhook

I tried making changes in repo

Edit
Preview

Spaces
2
No wrap

```

1  name: Java CI
2
3  on: [push]
4
5  jobs:
6    build:
7
8      runs-on: ubuntu-latest
9
10     steps:
11       - uses: actions/checkout@v1
12       - name: Set up JDK 1.8
13         uses: actions/setup-java@v1
14         with:
15           java-version: 1.8
16       - name: Build with Maven
17         run: mvn package --file pom.xml
18       .....

```

Marketplace

Documentation

Search Marketplace for Action

Featured Actions

Cache
By actions
Cache artifacts like dep build outputs to improve execution time

Setup Node.js environ
By actions
Setup a Node.js environment problem matchers and downloading and adding

Upload a Build Artifact
By actions
Upload a build artifact by subsequent workflow

Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element

We can able to see that the job is built

Jenkins
job-1

Status
Changes
Workspace
Build Now
Configure
Delete Project
GitHub Hook Log

✓ job-1
webhook job

Permalinks

- Last build (#1), 1 min 20 sec ago
- Last stable build (#1), 1 min 20 sec ago
- Last successful build (#1), 1 min 20 sec ago
- Last completed build (#1), 1 min 20 sec ago



Jenkins

job-1

▼

Workspace of job-1 on Bu...

▼

Status

Changes

Workspace

Build Now

Configure

Delete Project

GitHub Hook Log

Rename

Workspace of job-1 on Built-In Node

job-1 / .git /

→

hooks

info

logs

objects

refs

config13-May-2026, 11:08:08 am245 B

description12-May-2026, 4:30:11 am73 B

FETCH_HEAD13-May-2026, 11:08:08 am117 B

HEAD13-May-2026, 11:08:08 am41 B

index13-May-2026, 11:08:08 am7.11 KiB

Build >

...



Jenkins

job-1

▼

#1

▼

Edit Build Information

Delete build '#1'

Polling Log

Timings

Git Build Data

building in workspace /var/lib/jenkins/workspace/job-1

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/job-1/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/manisha-a99/Techie_horizon_Login_app.git # timeout=10

Fetching upstream changes from https://github.com/manisha-a99/Techie_horizon_Login_app.git

> git --version # timeout=10

> git --version # 'git version 2.50.1'

> git fetch --tags --force --progress -- https://github.com/manisha-a99/Techie_horizon_Login_app.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/master^{commit} # timeout=10

Checking out Revision 40667569cd2b02fb93c2b5938bfe288b40b1a4f0 (refs/remotes/origin/master)

> git config core.sparsecheckout # timeout=10

> git checkout -f 40667569cd2b02fb93c2b5938bfe288b40b1a4f0 # timeout=10

Commit message: "Update Maven workflow to include additional steps"

First time build. Skipping changelog.

[job-1] \$ /bin/sh -xe /tmp/jenkins7667202507754264698.sh

+ echo 'webhook triggered successfully'

webhook triggered successfully

Finished: SUCCESS

3. ConfigurePoll SCManBuild Periodicallyoptions in the Jenkins job.

Go to the Jenkins and create a new item

New Item

Enter an item name

PollSCM

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

General

Enabled



Description

Job to check pollSCM build trigger

Plain text [Preview](#)

☐ Enable project-based security

☐ Discard old builds ?

Select git repo

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

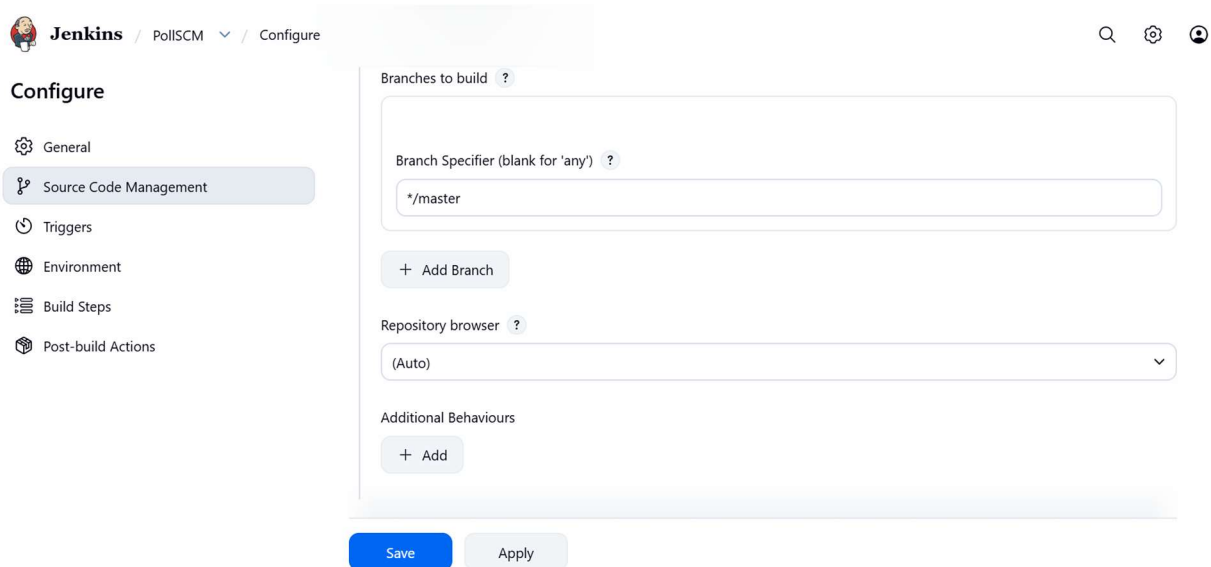
https://github.com/manisha-a99/Techie_horizon_Login_app.git

Credentials ?

- none -

+ Add

Advanced ▾



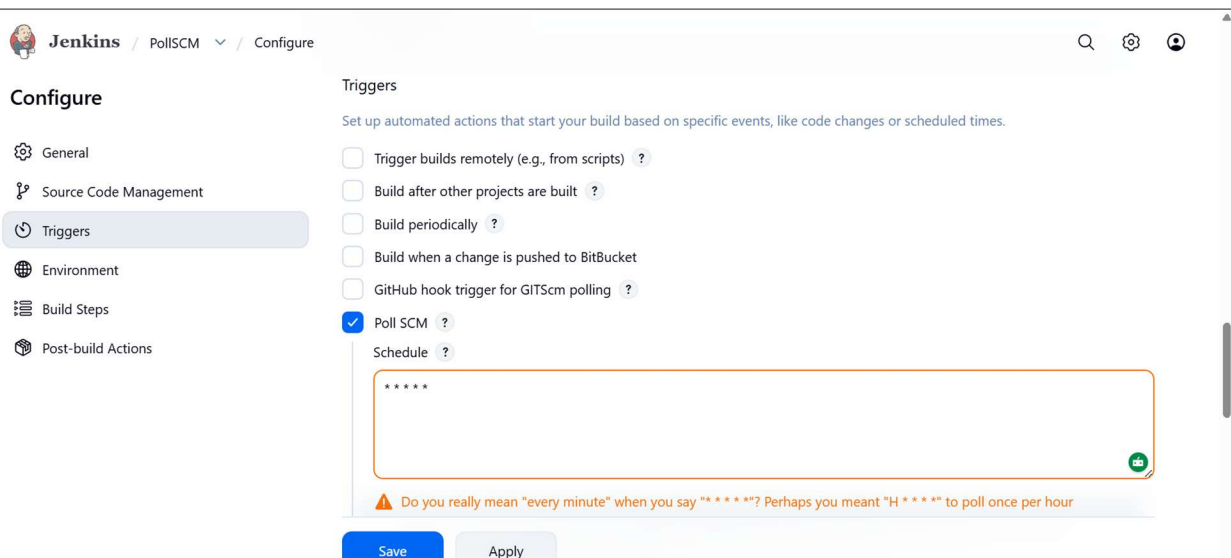
The image shows the Jenkins configuration page for PollSCM. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is titled 'Configure' and contains the following fields:

- Branches to build**: A text input field with the value `*/master` and a '+ Add Branch' button below it.
- Repository browser**: A dropdown menu with the value '(Auto)' and a downward arrow.
- Additional Behaviours**: A '+ Add' button.

At the bottom of the configuration area are two buttons: 'Save' (blue) and 'Apply' (grey).

Select pollSCM under triggers

Here we are scheduling for every min, every hour, every day. So that whenever we make any changes in repo then the job will run



The image shows the Jenkins configuration page for Triggers. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers (selected), Environment, Build Steps, and Post-build Actions. The main content area is titled 'Triggers' and contains the following fields:

- Set up automated actions that start your build based on specific events, like code changes or scheduled times.**
- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☐ Build when a change is pushed to BitBucket
- ☐ GitHub hook trigger for GITScm polling ?
- ☒ Poll SCM ?

Below the 'Poll SCM' checkbox is a 'Schedule' field with a text input containing `*****` and a green 'test' button. A warning message is displayed below the schedule field: **Do you really mean "every minute" when you say "*****"? Perhaps you meant "H*****" to poll once per hour**.

At the bottom of the configuration area are two buttons: 'Save' (blue) and 'Apply' (grey).

Now I am making changes in repo like I added a line

```
15 MSQl DB Installation Steps for Linux ubuntu 14.04:
16 - $ sudo apt-get update
17 - $ sudo apt-get install mysql-server
18
19 Then look for the file :
20 - /src/main/resources/accountsdb
21 - accountsdb.sql file is a mysql dump file.we have to import this dump to mysql db server
22 - > mysql -u <user_name> -p accounts < accountsdb.sql
23
24 - `Execute SonarQube Scanner` > `Analysis properties` (it is mandatory).
25 ```sh
26     sonar.projectKey=Sabear
27     sonar.projectName=Sabear
28     sonar.projectVersion=1.0
29     sonar.sources=/var/lib/jenkins/workspace/$JOB_NAME/src/
30     sonar.binaries=target/classes/com/visualpathit/account/controller/
31     sonar.junit.reportsPath=target/surefire-reports
32     sonar.jacoco.reportPath=target/jacoco.exec
33     sonar.java.binaries=src/main/java/com/visualpathit/account/
34     ...
35 java
36
37
```

Use **Control + Shift + m** to toggle the **tab** key moving focus. Alternatively, use **esc** then **tab** to move to the next interactive element on the page.

Here the build is triggered

 **Jenkins** / PollSCM #2

Status

</> Changes

Console Output

Edit Build Information

Delete build '#2'

Polling Log

Timings

Git Build Data

Previous Build

 **#2 (13-May-2026, 11:46:08 am)**

Add description

Keep this build for

Started by an SCM change

Started 24 s
Took 0.19 se

This run spent:

- 7.3 sec waiting;
- 0.19 sec build duration;
- 7.5 sec total from scheduled to completion.

 **Revision:** 5a4a075b764bf5551e26e11de5d67cb4c12befcb

Repository: https://github.com/manisha-a99/Techie_horizon_Login_app.git

- refs/remotes/origin/master

</>

Fix formatting issues in README.md 5a4a075

noreply at 11:45 am 13/05/26

 **Jenkins** / PollSCM #2

Console Output

Edit Build Information

Delete build '#2'

Polling Log

Timings

Git Build Data

Previous Build

Started by an SCM change

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/PollSCM

The recommended git tool is: NONE

No credentials specified

> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/PollSCM/.git # timeout=10

Fetching changes from the remote Git repository

> git config remote.origin.url https://github.com/manisha-a99/Techie_horizon_Login_app.git # timeout=10

Fetching upstream changes from https://github.com/manisha-a99/Techie_horizon_Login_app.git

> git --version # timeout=10

> git --version # 'git version 2.50.1'

> git fetch --tags --force --progress -- https://github.com/manisha-a99/Techie_horizon_Login_app.git +refs/heads/*:refs/remotes/origin/* # timeout=10

> git rev-parse refs/remotes/origin/master^{commit} # timeout=10

Checking out Revision 5a4a075b764bf5551e26e11de5d67cb4c12befcb (refs/remotes/origin/master)

> git config core.sparsecheckout # timeout=10

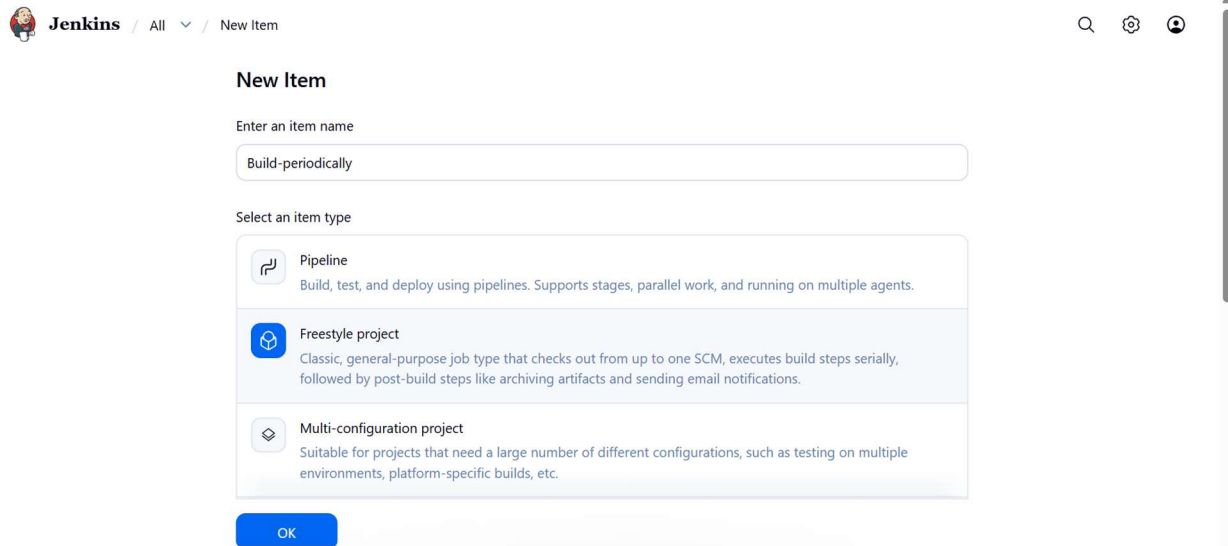
> git checkout -f 5a4a075b764bf5551e26e11de5d67cb4c12befcb # timeout=10

Commit message: "Fix formatting issues in README.md"

> git rev-list --no-walk 40667569cd2b02fb93c2b5938bfe288b40b1a4f0 # timeout=10

Finished: SUCCESS

Lets try build periodically



The image shows the Jenkins 'New Item' dialog. At the top, the Jenkins logo is followed by a breadcrumb 'All / New Item'. On the right, there are icons for search, settings, and a user profile. The main title is 'New Item'. Below it, there is a text input field labeled 'Enter an item name' containing the text 'Build-periodically'. Underneath is a section 'Select an item type' with three options: 'Pipeline' (described as 'Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.'), 'Freestyle project' (described as 'Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.'), and 'Multi-configuration project' (described as 'Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.'). The 'Freestyle project' option is selected and highlighted. At the bottom left is a blue 'OK' button.

Jenkins / All / New Item

New Item

Enter an item name

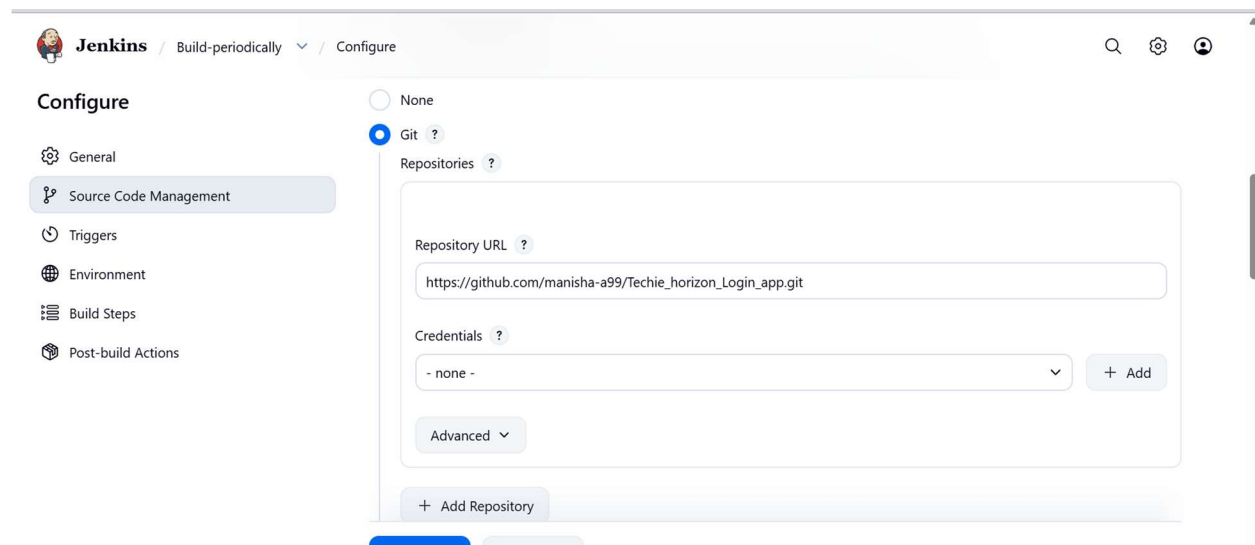
Build-periodically

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Select git repo



The image shows the Jenkins 'Configure' page for the 'Build-periodically' job. The breadcrumb is 'Jenkins / Build-periodically / Configure'. On the right, there are search, settings, and user profile icons. On the left, a sidebar titled 'Configure' lists several sections: 'General', 'Source Code Management' (which is highlighted), 'Triggers', 'Environment', 'Build Steps', and 'Post-build Actions'. The main content area shows the 'Git' option selected under 'Repositories'. Below this, there is a 'Repository URL' field containing 'https://github.com/manisha-a99/Techie_horizon_Login_app.git'. There is also a 'Credentials' dropdown menu currently showing '- none -' and an '+ Add' button. An 'Advanced' dropdown is visible below the credentials. At the bottom, there is a '+ Add Repository' button.

Jenkins / Build-periodically / Configure

Configure

- General
- Source Code Management**
- Triggers
- Environment
- Build Steps
- Post-build Actions

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/manisha-a99/Techie_horizon_Login_app.git

Credentials ?

- none -

+ Add

Advanced

+ Add Repository

Jenkins / Build-periodically / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Branches to build ?

Branch Specifier (blank for 'any') ?

*/master

+ Add Branch

Repository browser ?

(Auto)

Additional Behaviours

+ Add

Jenkins / Build-periodically / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☒ Build periodically ?

Schedule ?

/ * * * *

⚠ Do you really mean "every minute" when you say "*/1 * * * *"? Perhaps you meant "H * * * *" to poll once per hour
Would last have run at Wednesday, May 13, 2026, 12:07:00 PM Coordinated Universal Time; would next run at Wednesday, May 13, 2026, 12:08:00 PM Coordinated Universal Time.

Jenkins / Build-periodically

- Status
- Changes
- Workspace
- Build Now
- Configure
- Delete Project
- Rename

Build-periodically

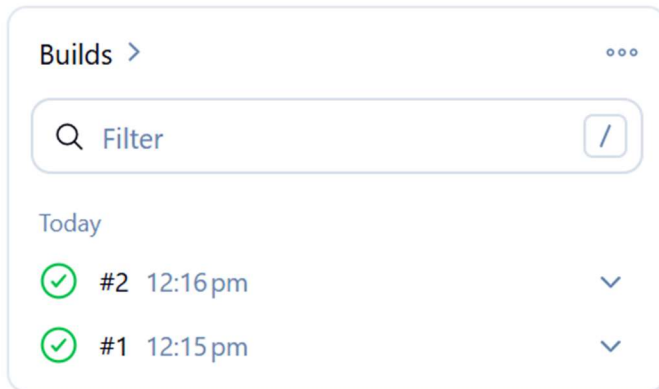
build periodically setup

Edit description

Permalinks

- Last build (#1), 15 sec ago
- Last stable build (#1), 15 sec ago
- Last successful build (#1), 15 sec ago
- Last completed build (#1), 15 sec ago

For every 1 minute the build is triggering



Build Periodically triggers builds based on time schedule, whereas Poll SCM checks source code repository for changes and triggers build only when changes are detected.

4. Create a bash script to take a backup of Jenkins server

Bash script to take backup of our server

```
root@ip-172-31-39-19:/home/ec2-user
#!/bin/bash

# Variables
BACKUP_DIR="/backup/jenkins"
JENKINS_HOME="/var/lib/jenkins"
DATE=$(date +%F-%H-%M-%S)

# Create backup directory if not exists
mkdir -p $BACKUP_DIR

echo "Starting Jenkins backup..."

# Create tar backup
tar -czvf $BACKUP_DIR/jenkins-backup-$DATE.tar.gz $JENKINS_HOME

echo "Backup completed successfully"

# Optional: Delete backups older than 7 days
find $BACKUP_DIR -type f -name "*.tar.gz" -mtime +7 -delete

echo "Old backups cleaned up"
~
~
~
-- INSERT --                                21.30
```



```

[root@ip-172-31-39-19 ~]# sudo chmod +x backup.sh
[root@ip-172-31-39-19 ~]# ./backup.sh
Starting Jenkins backup...
tar: Removing leading `/' from member names
/var/lib/jenkins/
/var/lib/jenkins/.java/
/var/lib/jenkins/.java/fonts/
/var/lib/jenkins/.java/fonts/21.0.11/
/var/lib/jenkins/.java/fonts/21.0.11/fcinfo-1-ip-172-31-39-19.us-east-2.compu
te.internal-amzn-2023-en-.properties
/var/lib/jenkins/secret.key
/var/lib/jenkins/secret.key.not-so-secret
/var/lib/jenkins/plugins/
/var/lib/jenkins/plugins/commons-lang3-api.jpi
/var/lib/jenkins/plugins/commons-lang3-api/
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/MANIFEST.MF
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/maven/
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/maven/io.jenkins.plugins
/
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/maven/io.jenkins.plugins
/commons-lang3-api/
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/maven/io.jenkins.plugins
/commons-lang3-api/pom.xml
/var/lib/jenkins/plugins/commons-lang3-api/META-INF/maven/io.jenkins.plugins

```

```

root@ip-172-31-39-19:~
/var/lib/jenkins/workspace/Build-periodically/src/test/java/com/visualpat
/account/controllerTest/UserControllerTest.java
/var/lib/jenkins/workspace/Build-periodically/src/test/java/com/visualpat
/account/modelTest/
/var/lib/jenkins/workspace/Build-periodically/src/test/java/com/visualpat
/account/modelTest/RoleTest.java
/var/lib/jenkins/workspace/Build-periodically/src/test/java/com/visualpat
/account/modelTest/UserTest.java
/var/lib/jenkins/workspace/Build-periodically/src/test/java/com/visualpat
/account/setup/
/var/lib/jenkins/workspace/Build-periodically/src/test/java/com/visualpat
/account/setup/StandaloneMvcTestViewResolver.java
/var/lib/jenkins/.config/
/var/lib/jenkins/.config/jgit/
/var/lib/jenkins/.config/jgit/config
/var/lib/jenkins/credentials.xml
/var/lib/jenkins/fingerprints/
/var/lib/jenkins/fingerprints/2d/
/var/lib/jenkins/fingerprints/2d/84/
/var/lib/jenkins/fingerprints/2d/84/2d1d5fcfda74bf84eca052cdb420.xml
/var/lib/jenkins/queue.xml.bak
/var/lib/jenkins/queue.xml
Backup completed successfully
Old backups cleaned up
[root@ip-172-31-39-19 ~]#

```

```

[root@ip-172-31-39-19 ~]# ls
backup.sh
[root@ip-172-31-39-19 ~]#

```

```

[root@ip-172-31-39-19 ~]# cd /backup/jenkins/
[root@ip-172-31-39-19 jenkins]# ls
jenkins-backup-2026-05-13-12-45-24.tar.gz

```

files in workspace folder

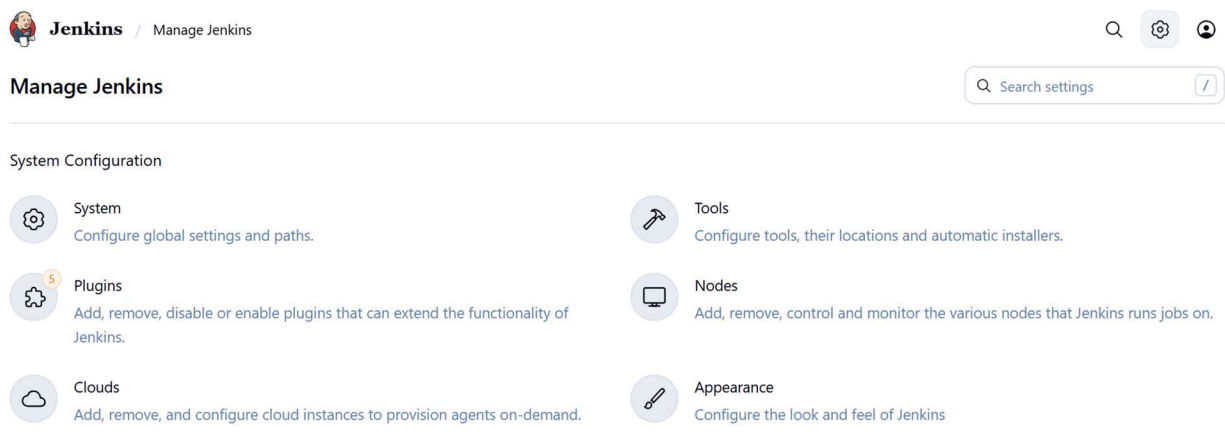
```

[root@ip-172-31-39-19 ~]# cd /var/lib/jenkins/
[root@ip-172-31-39-19 jenkins]# ls
config.xml
credentials.xml
fingerprints
hudson.model.UpdateCenter.xml
hudson.plugins.emailext.ExtendedEmailPublisher.xml
hudson.plugins.git.GitTool.xml
identity.key.enc
io.jenkins.plugins.prism.PrismConfiguration.xml
jenkins.install.InstallUtil.lastExecVersion
jenkins.install.UpgradeWizard.state
jenkins.model.JenkinsLocationConfiguration.xml
jenkins.plugins.git.GitHooksConfiguration.xml
jenkins.security.QueueItemAuthenticatorConfiguration.xml
jenkins.security.UpdateSiteWarningsConfiguration.xml
jenkins.security.apitoken.ApiTokenPropertyConfiguration.xml
jenkins.telemetry.Correlator.xml
jobs
logs
nodeMonitors.xml
org.jenkinsci.plugins.gitclient.GitHostKeyVerificationConfiguration.xml
plugins
queue.xml
queue.xml.bak
scriptApproval.xml
secret.key
secret.key.not-so-secret
secrets
updates
userContent
users
workspace
[root@ip-172-31-39-19 jenkins]# cd workspace
[root@ip-172-31-39-19 workspace]# ls
Build-periodically Private-job dev job-1 private-build
PollSCM Private-job@tmp 'job -3' job-2 private-build@tmp
[root@ip-172-31-39-19 workspace]# |

```

5. Take a backup of Jenkins using the ThinBackup plugin.

Go to manage → plugins



The screenshot shows the Jenkins web interface. At the top, there's a header with the Jenkins logo, the text "Jenkins / Manage Jenkins", and search, settings, and user icons. Below the header is a "Manage Jenkins" section with a search bar labeled "Search settings". The main content area is titled "System Configuration" and contains six configuration options, each with an icon and a description:

- System**: Configure global settings and paths.
- Tools**: Configure tools, their locations and automatic installers.
- Plugins**: Add, remove, disable or enable plugins that can extend the functionality of Jenkins. (This option is highlighted with a yellow background and a red "5" badge.)
- Nodes**: Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds**: Add, remove, and configure cloud instances to provision agents on-demand.
- Appearance**: Configure the look and feel of Jenkins.

Install the plugin

Jenkins / Manage Jenkins / Plugins

Plugins

Search: backup

Install Available plugins Installed plugins Advanced settings

Install	Name	Released	Health
☐	Backup 1.6.1 Miscellaneous Backup or restore your Hudson configuration files This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	14 yr ago	60
☒	ThinBackup 2.1.4 Miscellaneous Backups the most important global and job specific configuration files.	1 mo 2 days ago	100

Installed successfully

Jenkins / Manage Jenkins / Plugins

Plugins

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

ThinBackup ☒ Success

Loading plugin extensions ☒ Success

→ [Go back to the top page](#)
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

go to manage Jenkins → Click on system

Jenkins / Manage Jenkins

Manage Jenkins

Search settings

System Configuration

System
Configure global settings and paths.

Tools
Configure tools, their locations and automatic installers.

Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

Clouds
Add, remove, and configure cloud instances to provision agents on-demand.

Appearance
Configure the look and feel of Jenkins

Click on system → choose thinbackup configuration

 **Jenkins** / Manage Jenkins ▾ / System ▾

ThinBackup Configuration

Backup directory ?

Backup schedule for full backups ?

Backup schedule for differential backups ?

Max number of backup sets ?

Creating backup directory and scheduling it for every 2 mins means the backup will
Be taken every 2 mins

 **Jenkins** / Manage Jenkins ▾ / System ▾

ThinBackup Configuration

Backup directory ?

 The directory does not exist, but will be created before the first run.

Backup schedule for full backups ?

Backup schedule for differential backups ?

Max number of backup sets ?

 **Jenkins**

Manage Jenkins

System

☒ Backup build results ?

☐ Backup build archive ?

☐ Backup only builds marked to keep ?

☒ Backup userContent folder ?

☒ Backup next build number file ?

☒ Backup plugins archives ?

☐ Backup config-history folder ?

☒ Backup additional files ?

Files included in backup (regular expression)

☐ Clean up differential backups ?

☐ Move old backups to ZIP files ?

☒ Stop the backup as soon as an exception occurs in the file handling ?

The backup is now setup and we will create backup manually

Go to manage Jenkins → tools and actions → Thinbackup → click on backup

 **Jenkins**

Manage Jenkins

 **Manage Old Data**
Scrub configuration files to remove remnants from old plugins and earlier versions.

Tools and Actions

 **Reload Configuration from Disk**
Discard all the loaded data in memory and reload everything from file system.
Useful when you modified config files directly on disk.

 **Jenkins CLI**
Access/manage Jenkins from your shell, or from your script.

 **Script Console**
Executes arbitrary script for administration/trouble-shooting/diagnostics.

 **ThinBackup**
Backup your global and job specific configuration.

 **Prepare for Shutdown**
Stops executing new builds, so that the system can be eventually shut down safely.

 Jenkins

Manage Jenkins

ThinBackup

+ New Item

Build History

Project Relationship

Check File Fingerprint

ThinBackup

Settings are now integrated in global configuration.

Backup nowRestore

Build Queue

No builds in the queue.

Build Executor Status

0/2

Verify in Jenkins server

```
[root@ip-172-31-39-19 ~]# cd /var/lib/jenkins/thinbackup/
[root@ip-172-31-39-19 thinbackup]# ls
FULL-2026-05-13_13-22
[root@ip-172-31-39-19 thinbackup]#
```

6.Set up a new Jenkins server and restore the backup taken in task 4.

Setup a new Jenkins server

aws

Search

[Alt+S]

United States (Ohio)

Manisha A (142083399660)

Manisha A

EC2 > Instances > Launch an instance

Name and tags

Name

backup-jenkins-server

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images

RecentsQuick Start

Amazon LinuxmacOSUbuntuWindowsRed HatSUSE Linux

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11....read more

ami-0278a2977a50e13fc

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

CancelLaunch instance

aws

Search

[Alt+S]

United States (Ohio)

Manisha A

EC2 > Instances > Launch an instance

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-5' with the following rules:

Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11...read more

ami-0278a2977a50e13fc

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

```
manis@MANISHAREDDY MINGW64 ~/Downloads
$ ssh -i "Jenkins-1.pem" ec2-user@ec2-3-141-170-242.us-east-2.compute.amazonaws.com
The authenticity of host 'ec2-3-141-170-242.us-east-2.compute.amazonaws.com (3.141.170.242)' can't be established.
ED25519 key fingerprint is SHA256:AQpyFm3CqLm3AxLToYZm7p/gsvNIwn8kIOqqEdMUQ
Q.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-141-170-242.us-east-2.compute.amazonaws.com' (ED25519) to the list of known hosts.
```



```

      #_
    ~\  #####_
   ~\  #####\
  ~\  \###|
 ~\  \#|
  ~\  V~' ->
    ~~~
      ~.._
        _/_/
         _/m/'

```

```

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-40-164 ~]$ sudo su
[root@ip-172-31-40-164 ec2-user]#
```

Install Jenkins

```
root@ip-172-31-40-164:/home/ec2-user
[root@ip-172-31-40-164 ec2-user]# sudo yum update
Last metadata expiration check: 0:00:53 ago on Wed May 13 13:48:29 2026.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-40-164 ec2-user]# sudo wget -O /etc/yum.repos.d/jenkins.repo
\
https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-05-13 13:49:55-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.78.133, 2a04:4e42:83::
5
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.78.133|:443... connect
d.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/j 100%[=====>]      267  --.-KB/s    in 0s

2026-05-13 13:49:55 (16.2 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [2
/267]

[root@ip-172-31-40-164 ec2-user]# sudo rpm --import https://pkg.jenkins.io
rpm-stable/jenkins.io-2026.key
[root@ip-172-31-40-164 ec2-user]# sudo yum upgrade
Jenkins-stable          6.0 kB/s | 833 B      00:00
Jenkins-stable          42 kB/s | 1.6 kB      00:00
Importing GPG key 0x14ABFC68:
  Userid      : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
  From        : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Is this ok [y/N]: y
Jenkins-stable          13 kB/s | 1.9 kB      00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-40-164 ec2-user]# sudo yum install java-21-amazon-corretto
```

```
root@ip-172-31-40-164:/home/ec2-user
[root@ip-172-31-40-164 ec2-user]# sudo yum install java-21-amazon-corretto -y
Last metadata expiration check: 0:00:15 ago on Wed May 13 13:50:46 2026.
Dependencies resolved.
=====
Package                                Arch    Version                                Repository    Size
=====
Installing:
java-21-amazon-corretto                x86_64  1:21.0.11+10-1.amzn2023.1            amazonlinux  218 k
Installing dependencies:
alsa-lib                               x86_64  1.2.7.2-1.amzn2023.0.3                amazonlinux  503 k
cairo                                  x86_64  1.18.0-4.amzn2023.0.3                 amazonlinux  717 k
dejavu-sans-fonts                      noarch  2.37-16.amzn2023.0.2                  amazonlinux  1.3 M
dejavu-sans-mono-fonts                 noarch  2.37-16.amzn2023.0.2                  amazonlinux  467 k
dejavu-serif-fonts                    noarch  2.37-16.amzn2023.0.2                  amazonlinux  1.0 M
fontconfig                             x86_64  2.13.94-2.amzn2023.0.2                amazonlinux  273 k
fonts-filesystem                      noarch  1:2.0.5-12.amzn2023.0.2               amazonlinux  9.5 k
freetype                               x86_64  2.13.2-5.amzn2023.0.2                 amazonlinux  422 k
giflib                                 x86_64  5.2.1-9.amzn2023.0.3                  amazonlinux  48 k
google-noto-fonts-common               noarch  20240401-1.amzn2023.0.2               amazonlinux  17 k
google-noto-sans-vf-fonts              noarch  20240401-1.amzn2023.0.2               amazonlinux  593 k
graphite2                              x86_64  1.3.14-7.amzn2023.0.2                 amazonlinux  97 k
harfbuzz                               x86_64  7.0.0-2.amzn2023.0.2                  amazonlinux  873 k
java-21-amazon-corretto-headless      x86_64  1:21.0.11+10-1.amzn2023.1            amazonlinux  97 M
javapackages-filesystem               noarch  6.0.0-7.amzn2023.0.6                  amazonlinux  12 k
langpacks-core-font-en                noarch  3.0-21.amzn2023.0.4                   amazonlinux  10 k
libICE                                 x86_64  1.1.1-3.amzn2023.0.1                  amazonlinux  76 k
libSM                                  x86_64  1.2.4-3.amzn2023.0.1                  amazonlinux  45 k
libX11                                 x86_64  1.8.10-2.amzn2023.0.1                 amazonlinux  659 k
libX11-common                          noarch  1.8.10-2.amzn2023.0.1                 amazonlinux  147 k
libXau                                 x86_64  1.0.11-6.amzn2023.0.1                 amazonlinux  33 k
libXext                                x86_64  1.3.6-1.amzn2023.0.1                  amazonlinux  42 k
libXi                                  x86_64  1.8.2-1.amzn2023.0.1                  amazonlinux  42 k
```

```
root@ip-172-31-40-164:/home/ec2-user

libXext-1.3.6-1.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libpng-2:1.6.37-10.amzn2023.0.12.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@ip-172-31-40-164 ec2-user]# sudo yum install jenkins -y
Last metadata expiration check: 0:00:36 ago on Wed May 13 13:50:46 2026.
Dependencies resolved.
=====
Package                Architecture Version      Repository    Size
=====
Installing:
jenkins                noarch      2.555.1-1    jenkins       95 M
Transaction Summary
=====
Install 1 Package

Total download size: 95 M
Installed size: 96 M
Downloading Packages:
jenkins-2.555.1-1.noarch.rpm      34 MB/s | 95 MB  00:02
-----
Total                            34 MB/s | 95 MB  00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
```

Start jenkins


```

[root@ip-172-31-40-164 ec2-user]# sudo systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service
→ /usr/lib/systemd/system/jenkins.service.
[root@ip-172-31-40-164 ec2-user]# sudo systemctl start jenkins
[root@ip-172-31-40-164 ec2-user]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; press
   Active: active (running) since Wed 2026-05-13 13:52:16 UTC; 1min 37s ago
   Main PID: 26066 (java)
     Tasks: 37 (limit: 1014)
    Memory: 371.6M
       CPU: 20.622s
    CGroup: /system.slice/jenkins.service
            └─26066 /usr/bin/java -Djava.awt.headless=true -jar /usr/share>

May 13 13:52:11 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:11 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:11 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:11 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:16 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:16 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:16 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:16 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:16 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:16 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
May 13 13:52:21 ip-172-31-40-164.us-east-2.compute.internal jenkins[26066]:>
lines 1-20/20 (END)

```

Copy a pem key

```

root@ip-172-31-39-19:~
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEAs67uI310epOnAcxtEFgJervmjesdd9wi ofvsVAw43N64Ticd
rEGAXEm7iZeZuuyF547AXsWXqrHz9n1Yz+bSxa8KLZyVPVIX1mT8ftQ6Q6on7fef
uG9Z3AZzyNetaum8KNSFi6IKQn2pAML+YVbwIG0rUQXTyefMZ0m+ZNYvhn2tcEE
jysDUxsH8969e/5hfd/H/05kr6pKeFdC3vgpMxN8zhLdko9q4ip7mKK111DCxxk2
sFhtDGerAaoSea8dF18JtiPHebWxirAQfUn9ZMRRah7CtukaVoFap/wY6BFUZpu+
8ArgJPsbYoXqH0c+2Sg+D5T3uygxzA8b0WNKGWIDAQABaoIBADZ9G9ZDwyk40XJo
LcLBD87aYGY1IHCSnyYEIXyzYCMKn1WD4ym1g/Z4nZhyGpza0RiRX6DdxsqKwHLF
PEN/ufccx7dhPpij5nqrkS5c6XZPiWBosw7axaNHgh8gL8S2MMsq8BmSAA590Nxc
RYv7M0KCGJqxZE9cLc2EXKpzkwHy+uKUJzYN5jBbmQXo7mCfK+n7fhzDCGja5+ab
9ld7D671dNzy+CLK7mUenvi9wF9q6Jy3RNoXC380erc4BvKbLUafw/LLxFE76D5W
Uq+ZVpByli0KxhLg6GPs5fMJfZDWgMVDcCwppqe5Htwlnaw46tnxdXJ2YS3+8NiGW
tnLFG/ECgYEA65Vo4YgoT4uwtVBAf59Eme/SVEcGhc1lQ0Zfki6qh1BS5yXUmLoR
60a2MiQ8jor1xGx+FtegQwsZuj1oX1wvEuKxaHgdjVe5Xrn9lSGN+SU1a0TAYsZQ
BYRKi+OF2P79kvZ5tnyU+ZC88vQylCARXE8bQj59YUaRo+MjesX3Lp8CgYEAw0FV
GX4fAwwxuOv2zwnd9/eA2ZR3pVHcoSgt/B8c1k6at1/01K+ET2fDER7pAYu2u/69
NqeD3tELPKnxmj8XtFLBahP1SxLMdyGoybPXWsnw10cDmYx/sGorPlbF81Cj7zUv
pKI4uHCD2oLnm80P17fVERwgQwnmhbg6CuKB/wUCgYEA1ZwEHXt4EahgXscFtLKV
YX8zF3IBvQB/8SA+SC5IMMFkanLC8jjFRrA3Q//XZ28WJ6S4L74UcTzLGobVXKIz
jOZANpmzMhX43wuGA1MJiH9GDHzeezGWHwp2raEAHGIZSshyVCVib2ft693AgEhH
dYvFC6vdw/ZY2GF8KL1kA8ECgYEA35RFpCdHwflQugS9NRNRkZwsTkYMKbbPp32
dt0f9h7AQIEmdOMscJI6aXD2zETfAS18t1ro06PgDa5RUonUTGb1IHq/xDxCofEX
fYL+0002yUEyoxf8zHsgUu97C643KhELuVV01ZGLcSkM25A75Q99mdtFwowEhDwx
mYbsgGECgYEAzrAL5Ju/ZsS4NRwa03oJts6K5e4w/q7FBgC0DyfUywsR1ZVDMNHJ
671FMkXShuzS2LzLeju/B3FdUwhdTfMFZ1YkDN0yLLLGb71CFIjDTVXA9U9D4Uc/
IGiwjGflc+TJFDqabAIAj3ewJBS2OhGSy/v/pqgfdPBZWvslqNjJGo=
-----END RSA PRIVATE KEY-----
~
~
~
~
~
~
~

```

```
Connection closed
[root@ip-172-31-39-19 ~]# chmod 400 test.pem
[root@ip-172-31-39-19 ~]# scp -i test.pem /backup/jenkins/jenkins-backup-2026-05-13-12-45-24.tar.gz ec2-user@3.136.160.153:/tmp
jenkins-backup-2026-05-13-12-45-24.tar.gz 100% 151MB 332.3MB/s 00:00
[root@ip-172-31-39-19 ~]# |
```

Checked in backup-jenkin-server whether the backup is done

```
[ec2-user@ip-172-31-40-164 ~]$ cd /tmp
[ec2-user@ip-172-31-40-164 tmp]$ ls
hsperfdata_jenkins
jenkins-backup-2026-05-13-12-45-24.tar.gz
systemd-private-64d93fb06b7c452fa244b1418a87222d-chronyd.service-7u7dwd
systemd-private-64d93fb06b7c452fa244b1418a87222d-dbus-broker.service-Y9FpJS
systemd-private-64d93fb06b7c452fa244b1418a87222d-policy-routes@ens5.service-208KId
systemd-private-64d93fb06b7c452fa244b1418a87222d-systemd-logind.service-AA9mYU
systemd-private-64d93fb06b7c452fa244b1418a87222d-systemd-resolved.service-UIEg7j
[ec2-user@ip-172-31-40-164 tmp]$ |
```